

Lecture 3: Execution Machinery

Rahul Krishna

Columbia University, COMS E6998-019, Fall 2026

Draft of September 25, 2026, for the meeting of September 25, 2026

Abstract

These notes explain how an agent’s runtime connects successive computations. We use extended state machines to describe state, transitions, and execution traces, then examine how graphs and events implement those rules. The discussion covers state updates, routing, scheduling, tool contracts, and execution authority. It also explains how budgets constrain execution and how controlled measurements support comparisons between implementations.

Contents

1 Agents as state machines	1
1.1 Why a state machine describes an agent	1
1.2 Extended state machines	2
1.3 Transition notation	2
1.4 Behaviours, traces and computation trees	2
1.5 The planning agent as an extended state machine	3
1.6 Reading	4
2 What connects one step to the next	4
2.1 From the EFSM variables to the state record	4
2.2 Model context, runtime state, durable state	5
2.3 Reducers and the information retained between steps	5
2.4 Field lifetimes	6
2.5 Invariants	6
2.6 State schemas of the loop and the planning agent	7
2.7 Reading	8
3 Routing, feedback, and stopping	8
3.1 Routing functions and guards	8
3.2 The same graph, different paths	8
3.3 Feedback edges	9
3.4 Loop bounds and termination	9
3.5 What a trace establishes	9
3.6 The demo: recorded endings and repairs	10
3.7 Reading	11
4 Scheduling, tools, and execution authority	11
4.1 Scheduling: when an update becomes visible	11
4.2 The tool contract	11
4.3 The dispatch cycle	11
4.4 Evidence of a request and its execution	12
4.5 Execution authority	12
4.6 The demo: scheduling and request refusal	13
4.7 Reading	13
5 Event-triggered realization	13
5.1 Events, handlers and inversion of control	13
5.2 An event-triggered version of the machine	14
5.3 Two frameworks	14
5.4 The demo: the planning agent as a LlamaIndex workflow	15
5.5 Reading	15
6 Budgets and controlled measurement	16
6.1 Admission and accounting	16
6.2 A budget invariant	17
6.3 Exhaustion and truncation	18
6.4 Controlled measurement	18
6.5 Reading	18

A Conceptual exercises

A.1 How an agent moves through a task	20
A.2 What an agent remembers between steps	21
A.3 Choosing the next action and knowing when to stop	21
A.4 Tool requests, execution, and access	22
A.5 Events and queued work	23
A.6 Budgets and spending	24

B Answers and explanations

B.1 How an agent moves through a task	26
B.2 What an agent remembers between steps	26
B.3 Choosing the next action and knowing when to stop	27
B.4 Tool requests, execution, and access	28
B.5 Events and queued work	29
B.6 Budgets and spending	29

1 Agents as state machines

1.1 Why a state machine describes an agent

A state machine is a simplified description of an agent. It omits the prompts, the model and the tool code so that we can examine execution without describing the whole program.

An agent runs in discrete steps, such as model calls, tool calls and checks. Between steps, it retains information and may receive new inputs. The program determines which data changes and continuations are permitted. We can therefore describe the agent as a state machine with i) a set of states, ii) inputs received during a step, iii) outputs emitted by a step, iv) an initial configuration, and v) a transition rule that maps the current state and an input to a next state and an output. A workflow intended to terminate also specifies terminal configurations, from which no further step is scheduled.

Definition 1.1 (State). State: “condition of the system at a particular point in time”. It “[e]ncodes everything about the past that influences the system’s reaction to current input” [18, slide 10].

Here, state contains the information the program retains between steps. It needs to include only the past information that affects the next step. A controller’s next action depends on its instructions and earlier observations. The runtime’s next decision may depend on how many model calls have been used. When the program retains both, two visits to the same controller can have different successors: the code is the same, but the data it reads differ.

The system boundary determines what counts as an input. Model replies and tool results arrive during a step, and the program cannot compute them from its retained data. We can represent workspace contents in two ways: include the files in the modelled environment, or treat a tool’s report about them as an input observation. Under either choice, **the record determines the permitted transitions. It does not determine which result arrives.**

Definition 1.2 (Determinism). Determinism: “In every state, for all input values, exactly one (possibly implicit) transition is enabled” [18, slide 29].

Whether the agent is deterministic depends on the system boundary. If each model reply is an input, routing is deterministic whenever every routing rule is a function of the merged record. The record enables exactly one outgoing edge, so the same record

and reply always select the same edge. If the model is inside the system and its reply is an unobserved internal choice, one record can lead to either a tool call or a submission. This description is nondeterministic. It uses nondeterminism in the standard way to represent unknown aspects of a system or its environment.

Neither reading assigns a probability to a reply. A probabilistic machine attaches a probability to each transition, whereas for a nondeterministic one “no such probability is known. We just know that any of the enabled transitions from a state can be taken” [18, slides 33 and 37]. Our descriptions identify possible continuations without assigning their likelihoods. Deterministic routing alone does not make a run reproducible. Replay requires the same inputs, including model replies that we do not control.

1.2 Extended state machines

An agent has a small number of control locations and a large amount of data. In a plain state machine, each distinct conversation and counter value would require a separate state. Such a diagram would make the control structure difficult to see.

Definition 1.3 (Extended state machine). “Extended state machines augment the FSM model with variables that may be read or written” [19, slide 4].

Definition 1.4 (Configuration). A configuration of an extended state machine is a pair (q, v) : the current control location q , shown as a node in the diagram, and a valuation v that assigns a value to every declared variable.

An agent’s control locations name its computations: plan, call the controller, run the tools, check a candidate, and end. Its variables hold the conversation, observations, plan, counters and status. In our agent, `(controller, model_calls=2, ...)` and `(controller, model_calls=5, ...)` are two configurations at one location. Both run the same code with different data. Drawing a separate node for each conversation would obscure this shared control structure.

Finite control does not imply a finite configuration space. An extended state machine for a traffic light at a pedestrian crossing has four locations and one variable, *count*, whose range is $\{0, \dots, 60\}$. It therefore has at most $4 \times 61 = 244$ configurations. A message list without a length bound has no such limit. **The control locations are finite; the configurations need not be.** Enumerating reachable configurations, as in safety analysis of a finite machine, may then no longer terminate. The set of locations and the guards between them remains finite.

1.3 Transition notation

An edge of an extended state machine can specify a guard, an output action and a set action. The guard determines whether the transition is enabled; the output action specifies what it emits; the set action updates the variables. We write *guard* / *output action* above the edge and the set action below it. For example, the traffic-light controller’s red-to-green edge has *count* ≥ 60 / *sigG* above *count* $:= 0$. The initial arrow carries the set action that initializes the variables. We declare variables, inputs and outputs separately, following the notation’s convention [19, slides 5–6]: “We make explicit declarations of variables, inputs, and outputs to help distinguish the three.”

In prose, we use a one-line shorthand. A transition from configuration (q, v) reads input e , tests guard g , emits output *out* and applies set action *set*:

$$(q, v) \xrightarrow{e; g / out; set} (q', v').$$

The guard is a predicate over the variables and input, $g(v, e)$. The

set action changes v to v' . Table 1 explains each element for an agent.

Element	Symbol	In an agent
Control location	q	The computation ready to run: a model call, a tool dispatch, a check. There are finitely many, each shown as one node.
Variables	v	What the program retains between steps: messages, observations, a plan, counters, check results, a status.
Input	e	A result that arrives during a step: a model reply, a tool result, a check result.
Guard	$g(v, e)$	A predicate over variables and input, such as <i>the reply requests a tool</i> or <i>the call budget is spent</i> .
Output action	<i>out</i>	A request to perform an operation, or a reported terminal outcome.
Set action	<i>set</i>	Store a reply, extend the observations, increment a counter, replace a plan.
Initial configuration	(q_0, v_0)	The first location, with the initial set action: opening messages present, counters at zero, no observations, status running.
Terminal configuration		A configuration from which no further computation is scheduled; the record carries the reason for ending.

Table 1: The elements of an extended state machine, interpreted for an agent. Adapted from the general notation for extended state machines.

Consider a transition from the controller. The call counter c was below the call budget B , so the controller could issue a model call. Its reply e contains a tool request. The reply is the input, the guard tests for tool calls, and the output action requests their execution. The set action appends the reply to the messages H and increments c . The next location is *tools*:

$$(controller, v) \xrightarrow{e; calls(e) \neq \emptyset / request(e); H := H \cdot e, c := c + 1} (tools, v').$$

When the tool results arrive, a transition from *tools* appends them as observations and can return control to the controller. This is the same location as before, with a different configuration: H, c and the observations have grown. Updating the variables allows the machine to make progress while revisiting a location.

1.4 Behaviours, traces and computation trees

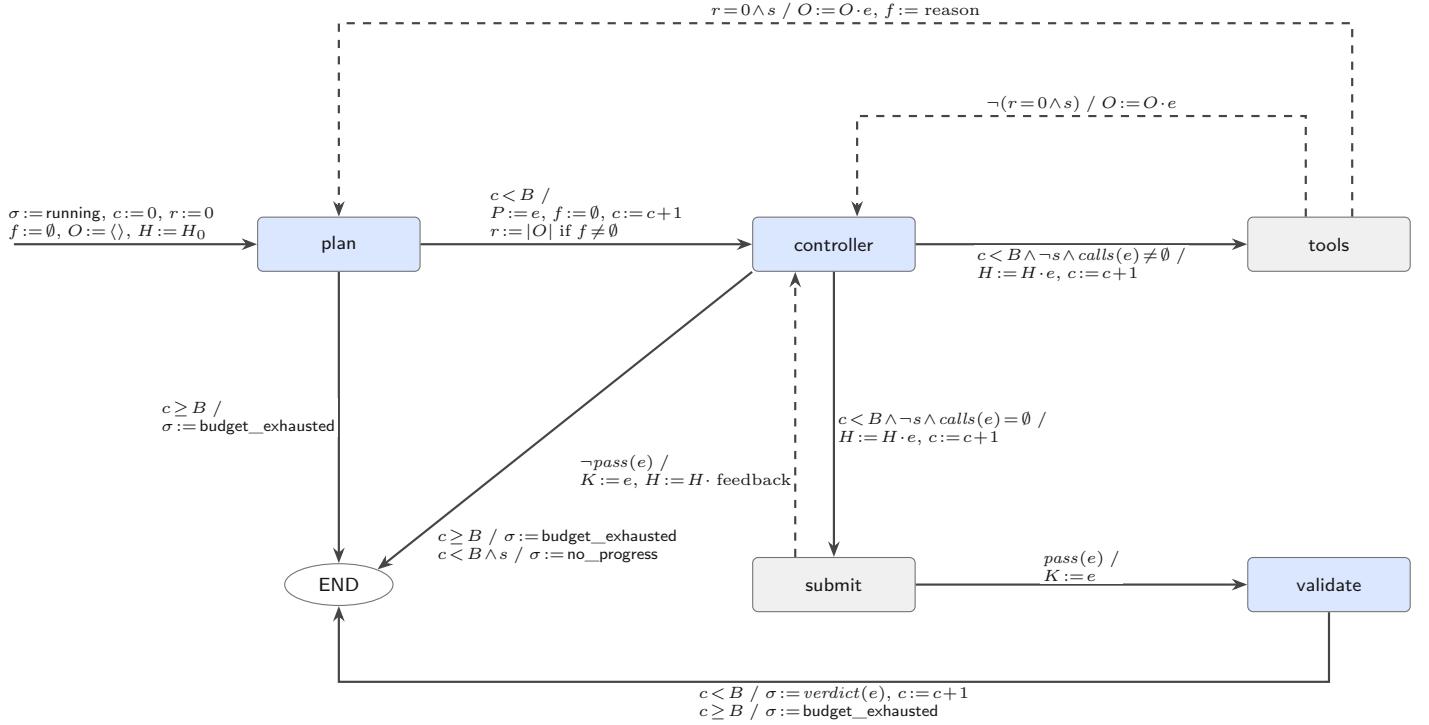
A transition specifies one permitted step. A run is a sequence of these steps. Three definitions describe what a run did or could have done [19, slide 8].

Definition 1.5 (Behaviour). “FSM behavior is a sequence of (non-stuttering) steps”.

Definition 1.6 (Trace). “A trace is the record of inputs, states, and outputs in a behavior”.

Definition 1.7 (Computation tree). “A computation tree is a graphical representation of all possible traces”.

A stuttering step occurs when inputs are absent, changes no state and emits no output. A behaviour omits these steps. Determinism determines whether a system has one behaviour or many. For a



Symbol	Meaning	Symbol	Meaning
e	The step's input: a reply, tool results, or check results.	O	Observations.
σ	Status.	P	Plan.
c	Model calls completed.	K	Check results.
B	Call budget (40).	f	Failed step (failed_step).
H	Messages.	r	Number of observations at the last revision (replanned_at).
H_0	The opening messages.	s	The last three observations since index r , including any just appended, are identical, and none is a successful edit.
$calls(e)$	The tool calls in reply e .	$pass(e)$	All three checks pass.
$verdict(e)$	accepted or rejected.		

Figure 1: The planning agent as an extended state machine, drawn from its `graph.py`. Each edge is labelled *guard / set action* in the notation of Section 1.3; output actions are omitted. Labels are stacked where two transitions share an arrow. Blue locations call a model; grey locations run tools or checks. The table under the drawing defines every symbol on the edges.

fixed input sequence, “A deterministic system exhibits a single behavior”, while “A non-deterministic system exhibits a set of behaviors” [18, slides 29 and 36]. The computation tree represents that set, branching wherever more than one transition is enabled.

A trajectory is one path through the computation tree, from the root to a terminal configuration or to the point where the run stopped. The tree unfolds the machine: each visit to a location appears as a new node. Four visits to the controller therefore produce four tree nodes, each with its own configuration and possible continuations. The machine diagram shows the permitted transitions. The tree shows the possible sequences of transitions from the initial configuration.

An agent’s event log is a practical trace that can survive the process. During execution, in-memory state holds the run’s current information. When the process exits, unsaved state is lost, while saved logs and workspace artifacts can remain. **State records what the run currently knows; the trace records what the run did.** Our agent writes six kinds of event: `tool_request`, `tool_result`, `model_call`, `routing`, `state_change` and `terminal`.

An event log meets Definition 1.6 only partly. It contains the inputs and outputs selected for logging and usually little state, because copying every variable at every step is expensive. Reconstructing a trajectory therefore combines logged facts with inferences from transition rules. A claim about a run needs to identify which of these supports it.

1.5 The planning agent as an extended state machine

Our running example is the *planning agent*, the complete workflow. The *planner* is its plan node and model role. The agent receives one scanner alert and drafts a plan. A reactive loop of controller and tool calls carries out the work. The agent submits a candidate patch to three mechanical checks, then a validator decides whether to accept it. In Figure 1, one transition represents completing a computation, applying its set action and selecting the next location. It omits the internal steps of model and tool calls. The figures and runs describe the planning agent when both runs were recorded. The current code adds two runtime nodes, `scan` and `normalize`, before `plan`.

The initial arrow enters `plan` with $\sigma := \text{running}$, no model calls, no observations, no revision ($r = 0$, $f = \emptyset$), and the opening system

and task messages H_0 . Figure 1 shows the following routes.

- **Planning.** `plan` makes one model call, stores the proposed plan in P , and moves to `controller`.
- **Acting.** A controller reply with tool calls, $calls(e) \neq \emptyset$, routes to `tools`, which appends the results to O . Unless the stall guard holds, control returns to `controller`.
- **Revising.** The stall guard is $r = 0 \wedge s$: the run has not revised its plan, and the last three observations are identical with no successful edit among them. When it holds, `tools` sets the handoff variable f (`failed_step`) and control returns to `plan`, which replaces P , clears f , and sets $r := |O|$ (`replanned_at`). From then on, s reads only the observations made since the revision.
- **Submitting.** A reply without tool calls, $calls(e) = \emptyset$, routes to `submit`, which runs the three checks: the patch applies, it touches the flagged line, and a rescan is clean. On $\neg pass(e)$ it appends feedback to H and returns to `controller`; on $pass(e)$ it moves to `validate`.
- **Judging.** `validate` makes one model call and always moves to `END`, with $\sigma := verdict(e)$, either `accepted` or `rejected`. A reply without a decision is recorded as a rejection made by the runtime.
- **Exhaustion.** `plan`, `controller` and `validate` each test $c < B$ before calling a model, with $B = 40$; when the test fails, the edge to `END` sets $\sigma := budget_exhausted$. At the controller, s can hold only after a revision, and it ends the run with $\sigma := no_progress$. The routing function after `submit` also checks the status, but no step on the way to `submit` can change it.

The planner stores its list of subgoals in a variable. Adding a subgoal does not add a control location, and routing does not test which subgoal is current. **A plan is data in the machine; its subgoals do not define control locations.** The controller receives the plan as a message. The plan affects the machine’s behaviour only through the controller’s replies.

Our worked run, `bc284c6b`, repairs a Bandit B608 alert (SQL injection) at line 46 of `testcode/BenchmarkTest00283.py`. Its log records this trajectory, with the tool each `tools` step ran:

```
plan
-> controller -> tools(read_file)
-> controller -> tools(search)
-> controller -> tools(read_file)
-> controller -> tools(edit)
-> controller -> submit -> validate -> END(accepted)
```

The trajectory contains 25 events. Events 1–2 record the model call in `plan` and the plan it stored. Events 3–18 record four tool rounds. Each contains a controller call, routing decision, tool request and tool result. The requests are `read_file` on the flagged file, `search` in `helpers/db_sqlite.py`, `read_file` on that helper, and `edit` on the flagged file. Events 19–22 are submission: a controller reply with no tool calls, the route to `submit`, three passed checks, and the route to `validate`. Events 23–25 record validation and the terminal event, `accepted`. The run made seven model calls: one by `plan`, five by `controller` and one by `validate`. It logged 12,693 input and 3,755 output tokens.

Figure 2 unfolds the beginning and end of this trajectory into a tree. Each visit to `controller` creates a new node whose reply selects `tools` or `submit`. At `submit`, the check result selects the branch; at `validate`, the verdict does. An exhaustion branch appears only where its guard permits it. On the recorded path, c never exceeds 7, so reaching `END` with `budget_exhausted` would require further rounds. The figure shows branches selected by replies or check results. Tool results can also change the route through the stall guard; those branches are omitted.

The log contains selected facts. A successful `tool_result` records success and result length, so the observations themselves are

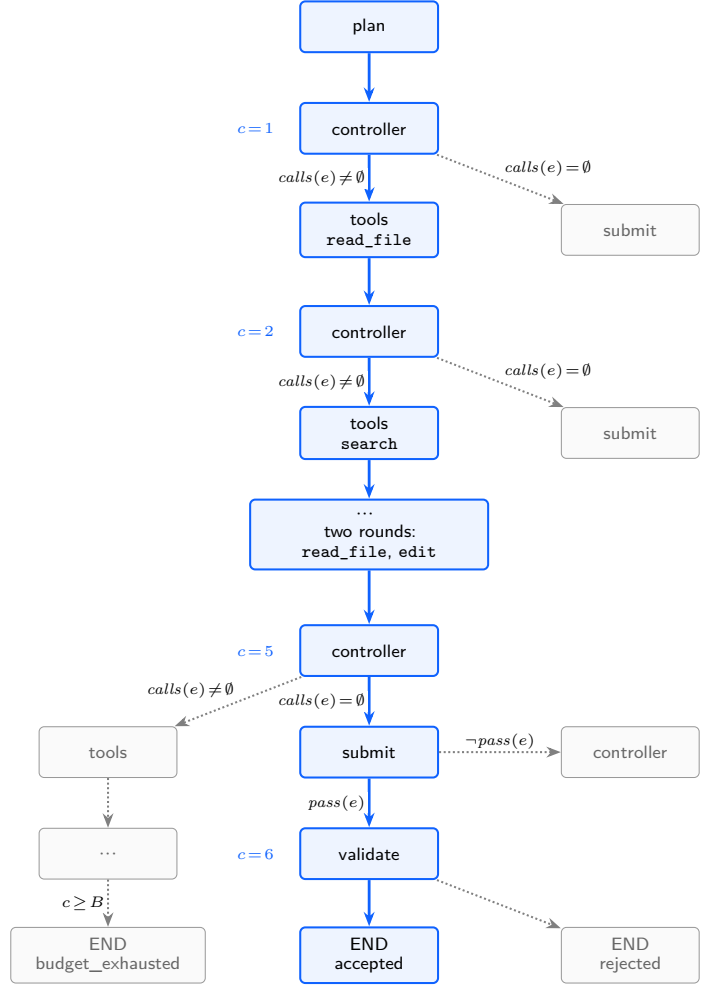


Figure 2: A fragment of the planning agent’s computation tree around run `bc284c6b`. Blue nodes and edges show the recorded trajectory; the `...` node omits two controller and tool rounds. Dotted grey branches are continuations of the guards permit under other replies or check results; the grey `...` omits further rounds. c is the number of model calls completed on reaching a node, derived from the log and the set actions. No probabilities are implied.

absent. The routes from `tools` to `controller` and from `plan` to `controller` produce no routing event. We infer these routes because the next event is a controller call and the transition rules allow no other successor. The counters are also absent from the log. Reconstructing this trajectory therefore requires both the log and the transition rules. Recovering every variable value at every step would require more detailed records.

1.6 Reading

Seshia [18], slides 10–37, on state, transitions and guards, determinism and receptiveness, and nondeterministic machines and their computation trees; and Seshia [19], slides 4–8, on extended state machines, their notation, the traffic-light example, and behaviours and traces.

Looking ahead. The machine has fixed control locations and variables whose values change at each step. The next section describes the record that holds those variables, the steps that may update each field, and the rules for merging updates with stored values.

2 What connects one step to the next

2.1 From the EFSM variables to the state record

A machine diagram shows the permitted control flow. A shared record supplies the information passed between steps. Routing

and dispatch, described in later sections, both read this record.

A configuration of an extended state machine pairs a control location with variable values ([Definition 1.4](#)). An agent’s runtime represents these separately. The application record holds the variables; the runtime tracks the control location, or which computation runs next. The *state schema* declares each variable’s name and type. Every step reads these variables and proposes updates to some of them. In our agent, the constructor call `StateGraph(AgentState)` constructs a graph with `AgentState` as its schema.

We describe a step with two operations. A node reads the current state s_t and obtains an external result e_t , then returns a partial update u_t . The runtime merges the update into the record, key by key:

$$u_t = \text{node}(s_t, e_t), \quad s_{t+1} = \text{merge}(s_t, u_t).$$

Here e_t denotes a result obtained during the node’s execution, such as a model reply or tool observation. The node’s actual parameter is the state alone. A field omitted from the update keeps its value in s_{t+1} .

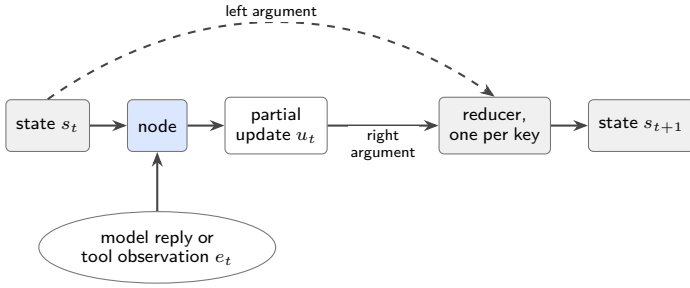


Figure 3: One transition. The node proposes an update, and the runtime merges it. Each key has a reducer whose left argument is the stored value and whose right argument is the value returned by the node.

A node returns only the keys it updates, because “the Node does not need to return the whole State schema - just an update” [9]. Our controller returns three of the loop’s ten keys: `messages`, `model_calls` and `usage`. On the two paths that stop the run before a model call, it returns only `status` ([Figure 4](#)).

The runtime applies the merge rule. It calls one reducer per updated key and saves the result. A node can therefore return its contribution without implementing the choice between appending to a list and replacing its stored value.

2.2 Model context, runtime state, durable state

Model context and runtime state contain much of the same information. The model receives what it needs for the current call. The runtime also retains counters and other fields that control execution. Saving enough state to resume after the process ends makes that state durable.

Counters illustrate runtime information excluded from model context. Our loop refuses a model call when `model_calls` reaches forty. It also refuses a call when the last three observations are identical. Neither number is shown to the model. Enforcement depends on the runtime’s comparison before the call, regardless of whether the model can see the limit in its prompt.

If our agent’s process stops during a run, a new process cannot recover its in-memory state record. Both graphs are compiled without a checkpointer. Trace files and workspace artifacts may survive, but they do not provide a resumable runtime record. LangGraph supports compiling a graph with a checkpointer, which gives “agents short-term memory” and “lets LangGraph applications keep useful information beyond a single graph run” [10]. Our JSONL trace contains selected events ([Section 1.4](#)).

Record	What it holds, and who reads it
Model context	Instructions, selected history and observations assembled for one model call. An agent with several model roles builds a separate context for each. In ours, the controller receives the conversation; the plan node and validator receive inputs assembled for their roles.
Runtime state	The record retained between steps, including fields excluded from model calls. Nodes, routing functions and limit checks read it.
Durable state	State saved beyond the process lifetime in a form that supports resuming a run. Resumption also requires the matching external state, such as the workspace, and a record of pending work.

Table 2: Three kinds of record. Each call’s model context is derived from runtime state. Durable state is runtime state saved beyond the process lifetime.

The saved trace does not contain enough information to resume the run.

Here, *durable* means that another process can read the saved record. Resuming can still repeat work. What must be repeated depends on checkpoint and task boundaries; an interrupted node may restart and repeat effects or paid calls not captured at a recoverable boundary. Later units examine persistence and recovery. This lecture concerns the record needed to execute a run; restart introduces further requirements.

2.3 Reducers and the information retained between steps

A merge function determines whether a field accumulates a history or retains only its latest value. This choice determines which evidence later steps can read.

In an extended state machine, each transition carries a set action that assigns new values to variables (19, slide 5). We implement that set action in two parts: a step proposes an update, then the runtime combines it with the stored value.

Definition 2.1 (Partial update). A partial update is the result of one step: a mapping from some of the state’s fields to new values. A field the update does not name keeps its stored value.

Definition 2.2 (Merge function). A merge function belongs to one field. It takes the stored value and the partial update’s value as arguments, then returns the field’s next value. Without a merge function, the update replaces the stored value.

LangGraph calls this merge function a *reducer*, the term used throughout these notes. A reducer has the form `(current, update) -> new` and is attached to a field with `Annotated[T, fn]`. The runtime calls it whenever a node returns that key. LangGraph’s `add_messages` reducer appends messages and replaces messages whose identifiers are already stored. Other binary functions can also serve as reducers, including the standard library’s `operator.add` and functions we write ourselves.

Three merge behaviours cover most fields. Each determines what later steps can read.

- **Append.** Add the update to the end of the stored list, keeping all earlier values. This is needed when a later step examines a sequence, for example, to check whether the same result occurred three times in a row. A field containing only the latest value cannot support that check.

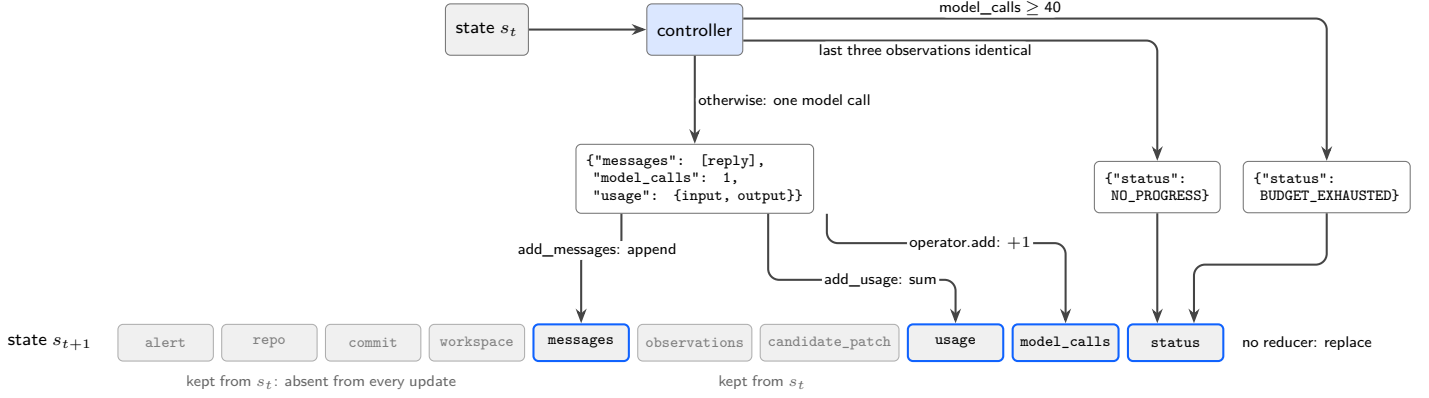


Figure 4: The controller’s three return sites against the loop’s ten-key record. The guard on each edge selects the path. The path that calls the model returns three keys, and each of the two paths that stop the run returns one. An arrow into a key names the reducer the runtime applies to it. `status` has no reducer, so the returned value replaces the stored one. The six keys without an arrow are absent from every update and keep their values from s_t .

- **Replace.** Overwrite the stored value with the update. This default behaviour suits a current plan, decision or status, where the new value supersedes the previous one and determines what later steps can use as evidence.
- **Accumulate.** Combine the update arithmetically with the stored value, usually by summing. Under this protocol, each node reports its own contribution, such as the cost of one call. The reducer maintains the running total from those deltas.

The reducer also determines whether an update can clear a field. For a list that appends updates, an empty update adds nothing: merging `[]` into `[x, y]` produces `[x, y]`. **An empty update leaves an appended field unchanged.** Clearing it requires bypassing the reducer and writing the new value directly, which LangGraph supports through **Overwrite**. Such a reset should produce a separate trace event because it bypasses the field’s ordinary merge rule.

2.4 Field lifetimes

A reducer defines how an update merges into a field. It does not define how long the resulting value remains valid. A stale value can mislead later steps.

A field’s *lifetime* is the part of a run during which its value remains valid evidence for its readers. It begins when the value is written and ends when an event supersedes or clears it. We choose the lifetime when designing the field. The runtime enforces only the reducer, so the nodes that write the field must follow its lifetime rules.

To choose a lifetime, identify who writes the field and which event makes its value invalid. These answers distinguish six lifetimes.

- **Identity.** Set once in the initial state and never updated by a node, so it needs no reducer. Use it for the task, target and workspace that define what the run concerns and where it may act. Updating an identity field changes the task during the run.
- **Run.** Valid throughout the run and only grows. A history uses an append reducer; a ledger uses an accumulating reducer. Use it when later steps or limits need the complete history, such as the conversation, observations, or calls and tokens spent.
- **Current.** Valid until its writer runs again, using the default replace rule. Use it for the latest plan, candidate, check result or status, when readers need the current value and the previous one is no longer valid evidence.
- **Handoff.** Written by one node for one reader, which clears it after acting. Use it when a step must explain why the next step was invoked, such as the reason for revising a plan. If the reader leaves it set, the next routing decision responds to the same signal again.
- **Window.** A span of run-lifetime history whose start is stored

in an index field. Use it when a check needs only the events since a given point, while other readers still need the full history. Replace the index to move the window; retain the history.

- **Attempt.** Valid for one attempt and cleared before a retry. Its reducer must allow this reset: a replaced field can receive its empty value, while an appended field needs **Overwrite**. Use it for results from a failed attempt that would mislead the next one, such as the candidate and its check results.

Counter lifetimes determine whether limits hold. A limit on attempts or model calls requires a run-lifetime counter that continues across attempts. **An attempt-lifetime counter resets after each failure, so it cannot enforce a limit across attempts.**

2.5 Invariants

A reducer or lifetime concerns one field. A property such as a spending limit constrains the whole run. Proving it requires considering every transition the machine can take.

Definition 2.3 (Invariant). “An invariant for a state machine is a predicate, P , on states, such that whenever $P(q)$ is true of a state, q , and $q \rightarrow r$ for some state, r , then $P(r)$ holds” [17, p. 5].

For an extended state machine, the states in this definition are configurations (q, v) , and the arrow denotes one transition. To establish a property of every reachable configuration by induction, we prove two statements: i) it holds initially, and ii) every transition preserves it. The second proof obligation covers every configuration satisfying the predicate, including unreachable ones. A property that holds in every reachable configuration may therefore need a stronger predicate before we can prove preservation by every transition.

Consider the property *model calls stay within the call budget*. Let B be the budget, c the recorded call counter, and k the number of completed model calls. The runtime can compare only c ; k describes the run’s history and is not a program variable. We want $k \leq B$ and prove the stronger predicate $P \equiv (c = k \wedge c \leq B)$. Initially, no call has been made and $c = 0$, so P holds. Transitions must then meet three conditions.

- **Admission** (Definition 6.1). Every transition that issues a model call has the guard $c < B$. Completing the call increases k by one, keeping it at most B .
- **Counting.** The update recording a completed call adds exactly one to c . An unreported call leaves c below k , allowing admission to exceed B . Reporting a call twice exhausts the budget early.

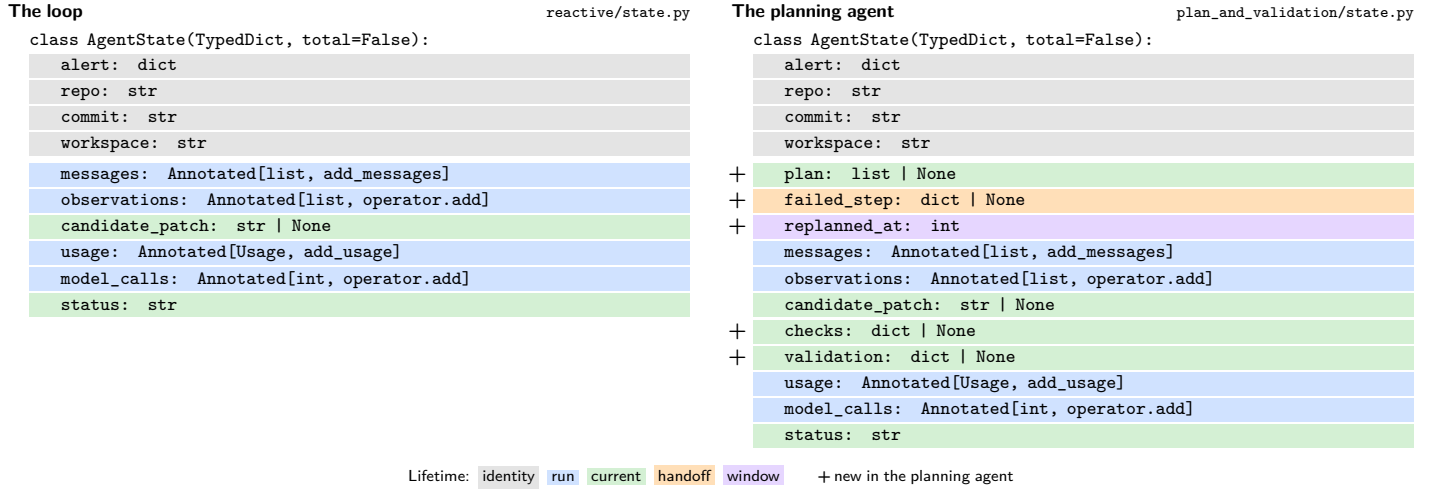


Figure 5: The two `AgentState` classes, with comments omitted. Shading indicates field lifetime (Section 2.4); + marks the planning agent’s five additional fields. Three use the current lifetime already present in the loop. `failed_step` and `replanned_at` introduce two further lifetimes. Neither class has an attempt-lifetime field, because neither agent retries.

- **Preservation.** A transition without a call leaves c unchanged. No transition, including plan revision, decreases it. This is the run-lifetime requirement from Section 2.4. Resetting c to zero preserves $c \leq B$ but breaks $c = k$, allowing later calls to exceed B .

The argument covers completed calls only. An interrupted call may incur cost without being counted, for example when an exception interrupts it. P does not cover that cost.

An accumulating reducer on c establishes part of the argument: merging adds the reported delta to the stored count. Nodes must still check the guard before calling and report exactly one call per completed call. **The reducer accumulates reported calls; each caller must report them correctly.** In our agent, c is `model_calls` and B is forty.

2.6 State schemas of the loop and the planning agent

Our state schema is the Python class `AgentState`, declared as a `TypedDict` (Figure 5). Each key names a record field, and its annotation specifies the value’s type. An annotation of `Annotated[T, fn]` also attaches the reducer `fn`. Setting `total=False` marks every key as optional for type checking. This declaration accommodates the partial updates that nodes return. `TypedDict` supplies declarations for type checkers [12]; it does not check returned values at run time.

The graph uses the class when `StateGraph(AgentState)` reads it during construction. It creates one channel per key, using the annotated reducer or replacement when none is specified. The remaining graph operations use those channels:

- `add_node(name, fn)` registers a node: a function that receives the whole state and returns a partial update.
- `add_edge(a, b)` fixes a transition from node `a` to node `b`.
- `add_conditional_edges(a, route, {...})` registers a routing function to run after `a`. It reads the updated state and returns the next node’s name without writing state.
- `compile()` checks the wiring and returns a runnable graph. `invoke(initial_state(...))` starts the run with a value for every key, so nodes do not read missing keys.

```

graph = StateGraph(AgentState)
graph.add_node("plan", self.plan)
graph.add_node("tools", self.tools)      # ... and three more
graph.add_edge(START, "plan")
graph.add_conditional_edges("tools", self.after_tools,
    {"controller": "controller", "plan": "plan"})
graph.add_edge("validate", END)
app = graph.compile()

```

The loop’s schema (Figure 5, left) uses three lifetimes. Identity fields name the scanner alert, fixture, pinned commit and workspace copy. Run-lifetime fields retain the information used by stall and budget checks: `messages` and `observations` append, while `model_calls` and `usage` accumulate. Our `add_usage` reducer sums four token counters separately, without a combined total. Cache reads cost a fraction of fresh input, and a combined total would hide an increase in cache reads alongside a decrease in fresh input. In both agents, the reply adapter populates only input and output counters. Both cache counters are zero in the planning agent’s two recorded runs. The current-lifetime fields, `candidate_patch` and `status`, hold the latest patch and execution status.

The planning agent (Figure 5, right) has five additional fields, each introduced with the node that writes it. The plan node writes `plan`, the submit node writes `checks`, and the validator writes `validation`. All three use the current lifetime because a new value supersedes the previous one. The other two fields use different lifetimes.

- **Handoff.** When the tools node detects a stall, it writes the reason to `failed_step`. The routing function reads this field and selects the plan node. That node reads the reason, revises the plan and clears `failed_step` to `None`. If it left the field set, the next pass through `tools` would return to the plan node again.
- **Window.** The `observations` list appends and is never cleared. After a revision, its last three entries are still the identical results that triggered revision. The plan node records the current observation count in `replanned_at`, and the stall check then reads only `observations[replanned_at:]`. This field also enforces the revision budget: a nonzero value indicates that revision has occurred, so a second stall ends the run.

Each field in the planning agent’s schema exists because a later computation reads what an earlier one wrote (Table 3).

Two code excerpts show how the call counter separates model context from runtime state (Section 2.2). The planning agent’s controller receives the conversation as context. Before every model call, admission compares the counter with the configured call budget. **Admission reads the call budget. The controller’s context does not contain it.** None of the planning agent’s prompts mentions the budget.

Fields	Why they persist and which steps read them
alert, repo, commit, workspace	Identify the task and its permitted workspace throughout the run. The workspace path names external state; the files are not in the record.
plan	The plan node writes it and adds a rendered message to the controller’s conversation. A revision reads the previous plan.
failed_step, replanned_at	The tools node records a stall; routing sends the run to the plan node, which clears failed_step and sets replanned_at to the revised plan’s observation boundary. Stall checks read from that index onward.
messages	The opening instructions, plan messages, controller replies, tool replies and check feedback form the controller’s conversation. This field excludes the replies of the plan node and validator.
observations	The tools node appends one structured result per tool call; the stall checks and the validator read them.
candidate_patch, checks	The submit node stores the patch and the check results, whether passed or failed. Routing tests the checks, and validation reads both.
validation, status	The validator stores its verdict, its reasons, and whether the model or runtime decided. status also records runtime stop reasons. Routing and the code that runs the graph and reports its outcome read this status.
model_calls, usage	Each of the three model nodes adds one call and that call’s tokens. Admission reads the call count; only the run’s closing summary reads the tokens.

Table 3: The planning agent’s fields, grouped by the node that writes them and the computations that read it.

```
# controller: the model receives the conversation
reply = self._call(model, state["messages"],
                  role="controller")

# admission: run before every model call
if state["model_calls"] >= self.config.budget:
    return {"status": TerminalStatus.BUDGET_EXHAUSTED}
```

We define a state graph in the following order.

1. Name the nodes and the routing decisions between them.
2. List the values that nodes and routing functions must read. These values define the schema. Omit fields that no computation reads.
3. For each field, identify its writer and the event that invalidates its value. Choose its lifetime using [Section 2.4](#).
4. Choose the reducer from the lifetime: none for identity, append or accumulate for run, and default replace for current, handoff and window. An attempt field needs a reducer that permits clearing.
5. Write the initial state so that every key has a value before the first node runs.
6. Write each node to return only its own keys. Add each schema field with the node that populates it.

2.7 Reading

LangChain [9], the implementation reference for state, schemas and reducers; and MIT 6.042 [17], §2 (p. 5), on invariants and the Invariant Theorem.

For reference while writing HW1: Lehtosalo [12] on what a `TypedDict` does and does not check at run time, and LangChain [10] on the checkpoints, threads and stores used for durability in [Table 2](#).

Looking ahead. A routing function reads the merged record, including the node’s latest update and previously stored values. It selects the next node or ends execution. The next section examines conditional edges, feedback to the plan node or controller, and three interpretations of a trace: recorded acceptance, a declared stop, and no recorded ending.

3 Routing, feedback, and stopping

3.1 Routing functions and guards

The routing function selects the next step by testing values in the record. The model helps produce those values; the function applies the transition rules written by the programmer.

Definition 3.1 (Conditional edge). A conditional edge is the set of outgoing transitions of one control location, together with a routing function that selects among them. The routing function reads the record after the location’s update has been merged, evaluates the guard of each outgoing transition, and returns the destination whose guard holds.

In the notation of [Section 1.3](#), a guard is a predicate $g(v, e)$ over the variables and the step’s input. The routing function receives these through the merged record $v' = \text{merge}(v, u)$. The node has already stored the information it retained from e . **A guard can test stored information; it cannot test information the node discarded.** Routing is deterministic in the sense of [Definition 1.2](#) when it selects exactly one destination for every merged record. A first-match chain with a final default does this by using test order to resolve overlapping conditions. The default may still be wrong; it also handles records the programmer did not anticipate.

A graph also needs a destination that ends the run. LangGraph provides it as **END**: “The END Node is a special node that represents a terminal node” [9]. A routing function that returns it stops the run. The record states the reason.

Three components have separate responsibilities.

- **Proposal.** The model emits a tool request, a plan, a candidate patch or a verdict. Each proposal is a value that the runtime must process before it can take effect.
- **Record.** The runtime merges the proposal into state through the reducers of [Section 2.3](#) and records what happened in the trace.
- **Decision.** The routing function reads named fields of the merged record and returns a destination according to the guards and limits written by the programmer.

In our planning agent, **after_controller** tests the status first. It returns **END** unless the run is still active. Otherwise, it returns **tools** if the last reply contains tool requests, or **submit** if it contains none. The function does not read the reply’s prose. A reply that says “the patch is correct, submit it” while requesting a tool therefore routes to **tools**.

A routing event records **by="model"** when the tested value came from model output. It records **by="runtime"** when that value came from a check, a limit, or a runtime default. **The field identifies the source of the tested value. It does not mean that the model chose the edge.** The programmer defined the edge and its guard.

3.2 The same graph, different paths

The compiled graph contains the locations, unconditional edges, and routing functions. Its structure stays the same across

runs. Runs can differ in their sequence of merged records and, consequently, in the destinations their routing functions select.

In the terms of Section 1.4, the compiled graph defines the machine, and each run follows one trajectory through its computation tree. Each visit to a location creates a new node in that tree, with its own configuration. A run that calls the controller twelve times passes through twelve branch points without changing the graph. **Compilation fixes the possible paths; the values encountered during a run determine its path.**

Figure 1 shows our planning agent’s machine, and Figure 2 unfolds part of its computation tree. Section 3.6 examines two recorded runs that follow different paths.

3.3 Feedback edges

A feedback edge returns control to a location the run has already visited. Its definition must specify the trigger and the information passed back. If it restarts work, it must also specify which fields retain their values and which are reset.

Agents of this kind commonly use two feedback edges with different triggers and destinations.

- **Revision.** A runtime check detects that progress has stalled and records the reason in a handoff field. The router then returns control to the planning location, which reads the reason, replaces the plan, and clears the field. A marker identifies the first observation under the revised plan, so later checks assess the revision using its own evidence (Section 2.4).
- **Repair.** When a proposed artifact fails a mechanical check, the runtime appends the check’s reasons to the conversation and the router returns the run to the location that proposed it. No fields need resetting. The conversation retains the failed proposal, and the next proposal replaces the stored one.

The first edge identifies a problem with the plan; the second identifies a problem with the artifact. Both continue the run and therefore need a bound. **A feedback edge without its own counter depends on the limits for the whole run.** Our planning agent has both edges. The first stall returns control to the plan node once. Failed checks return control and their reasons to the controller as often as the call budget allows.

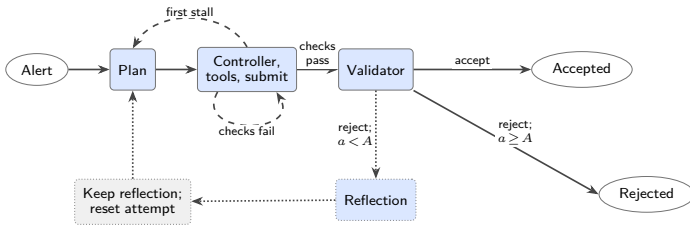


Figure 6: A proposed extension to the planning agent. Solid and dashed edges exist in the implementation; dotted nodes and edges show the extension. Blue boxes make model calls, the grey box performs runtime work, and ellipses mark the input and two endings. Dashed arrows indicate feedback. The middle box represents the controller, tools, and submit steps. a counts attempts and A is the attempt limit. In the implemented planning agent, every rejection ends the run.

The proposed extension adds a third feedback edge (Figure 6). In the current planning agent, the validator’s verdict always ends the run. The extension would send a rejection to a reflection step, which records what went wrong, then start a new attempt at the plan node. This edge requires an explicit reset policy. Results from the failed attempt, including its plan, patch, and checks, would mislead the next attempt. These fields therefore have the attempt lifetime defined in Section 2.4 and require explicit clearing. The reflection, attempt counter a , and call counter have the run lifetime and must survive the reset. **Resetting the attempt counter with each attempt prevents it from reaching its limit.**

3.4 Loop bounds and termination

An exit from a loop does not by itself guarantee termination. An acceptance predicate identifies a successful ending without ensuring that one occurs. If a run’s patches keep being rejected or are never submitted, another condition must stop it. A termination argument establishes that some condition will eventually do so.

Definition 3.2 (Variant). To show that a loop terminates: “Assuming the loop invariant holds at the start of each iteration, show that some quantity strictly decreases, and that it cannot decrease indefinitely. This quantity is called the decrementing function or loop variant.” [5]

For a graph, an iteration is a cycle that continues the run. We use a simple variant: a nonnegative integer computed from the record that strictly decreases on every continuing cycle. It cannot decrease indefinitely, so its initial value bounds the number of cycles. This argument does not bound the duration of a step.

Remaining calls, $B - c$, is one candidate. The call invariant of Section 2.5 keeps $c \leq B$, so the quantity is nonnegative. It serves as a variant under two conditions: i) every cycle includes a transition that consumes a call, and ii) every step between calls returns or times out. A cycle consisting only of runtime work consumes no call. Examples include a check that repeats itself or a tool retry that does not consult the model. In such cycles, $B - c$ stays constant. **A cycle that consumes no model calls needs its own bound.**

A stall check detects a particular pattern of non-progress, such as repeated identical results, and then ends or redirects the run. A run can fail to progress without producing that pattern. For example, each search may return a different irrelevant result. **Detecting one failure pattern does not prove termination.** ReAct’s recorded trajectories include repeated actions that make no progress [20, §3.3].

Three runtime checks can end a run before a verdict. Each compares values in the merged record.

- **Budget.** A limit on model calls, checked before each call rather than after. A limit checked afterwards cannot prevent the cost of the call. When the limit holds on every cycle, remaining calls is the variant.
- **Stall.** A check that recognizes repeated results. It can redirect the run to revision the first time and end it the next, which bounds the revisions but not the run.
- **Recursion limit.** LangGraph caps the number of supersteps (Definition 4.1) one execution may run and raises an error when execution reaches that cap [9]. This check uses no information about the task or its costs. It provides a failsafe against incorrect graph wiring.

In our planning agent, the call budget is forty calls, and the recursion limit is four times the call budget.

3.5 What a trace establishes

A trace can support three distinct claims: i) execution terminated; ii) the acceptance procedure passed; iii) the artifact satisfies the task’s postcondition. Each claim is stronger than the previous one. A terminal event establishes the first; an accepting terminal status also establishes the second. No event establishes the third. The acceptance procedure provides evidence about the artifact, without proving that it meets the postcondition. **The trace records the outcome of acceptance checks. It does not establish that the patch is correct.**

The trace writer uses a fixed vocabulary of five terminal statuses: accepted, rejected, budget_exhausted, no_progress and error.

Feedback edge	Trigger	Information passed back	What is reset	What bounds it
Revision: tools to plan node	Three identical observations within the current plan’s window, none a successful edit.	The stall reason and the previous plan, read by the plan node.	The plan is replaced, the handoff field cleared, and the stall window moved; observations are kept.	At most once per run. A second stall ends the run with no_progress .
Repair: submit to controller	At least one of the three checks fails.	The checks’ reasons, as a message in the conversation.	Nothing. The next submission replaces the stored patch and check results.	The call budget alone.
New attempt: validator to reflection to plan node (proposed, not implemented)	The validator rejects, and fewer than <i>A</i> attempts have been made.	A written reflection on the rejected attempt.	Every attempt-lifetime field, by explicit writes.	The attempt limit, on a counter with the run lifetime, and the call budget.

Table 4: The planning agent’s two implemented feedback edges and the proposed third edge. Only the first two rows describe current behavior.

It refuses other values. Closing a trace without a terminal event raises `NoTerminalStatus`. **A missing ending at normal trace closure indicates an instrumentation defect; it is not a reportable outcome.** When an exception escapes the run, the writer records terminal status **error**, then writes a record naming the exception. The status identifies the kind of ending; the next record identifies its cause.

If the process receives `SIGKILL`, no handler runs. The file ends at the last written event, without a terminal or exception record. A power loss may also lose that last event: the writer flushes events to the operating system but does not force them to disk (Table 5).

What the trace records	What it supports
A verdict, then a terminal event with status accepted	Execution terminated and the acceptance procedure passed. Whether the patch meets the task’s postcondition is a separate question the trace does not answer.
A terminal event with a stop status	Execution terminated without acceptance, for the stated reason, with whatever partial evidence the run had gathered. For error , the record that follows names the exception.
No terminal event	Termination has not been established by this record. The run may still be active, or its process may have been killed without running any handler, as a <code>SIGKILL</code> or a power loss would leave it.

Table 5: Three interpretations of a trace. The absence of a terminal event limits what the record establishes. Classifying that absence as **error** assigns a cause without evidence.

These interpretations depend on three distinctions.

- **Errors and outcomes.** A refused tool call is an observation inside a run that can still end **accepted**, while a process killed between two events leaves a run with no status at all.
- **Findings and status.** An assessment that reports insufficient information is a completed assessment, while a declared stop can carry partial evidence and no finding.
- **Reaching the end and the reason.** Reaching **END** establishes only that the graph stopped. The reason is in the status, which the runtime sets.

The runtime controls termination as well as routing. A model’s claim that it has finished is a proposal. The runtime applies completion rules and sets the status; the trace records that status.

3.6 The demo: recorded endings and repairs

Our planning agent implements these routing rules and limits. Its router after submission, `after_submit`, has three exits: **END** if the status has changed, `validate` if all three checks passed, and `controller` otherwise. The stall check fires on three consecutive identical observations, none of which is a successful edit. The first stall returns control to the plan node, which records `replanned_at`. A second stall, measured using only observations since that point, ends the run with **no_progress**. Neither recorded run revises its plan.

Remaining calls is a variant for the whole graph. Every cycle passes through the controller, which either makes a call or ends the run. The recursion limit cannot fire first: every step without a model call is followed by a step that makes a call or ends the run. A run therefore takes at most about twice the call budget in super-steps. The requirement that steps return or time out (Section 3.4) holds only in part. Rescanning and patch application have timeouts. Model calls and file tools depend on limits imposed by the client library and operating system; our configuration sets none.

The two recorded runs of the planning agent have different endings. One includes two repairs. The code also defines a third ending, budget exhaustion. The step numbers below are the trace files’ `step` fields.

- **Accepted.** Run `bc284c6b` ends **accepted** after one submission; Section 1.5 walks through its events.
- **Returned twice.** Run `28e37a00` submits at step 24. At step 25 the patch applies, but the edit does not touch the flagged line and the rescan still reports the finding. Step 26 routes back to `controller`, recording `by="runtime"` and the two reasons. The controller receives those reasons as a conversation message. After three tool rounds (steps 27–38), it submits again. At step 41 the rescan is clean, but the flagged-line check still fails. Step 42 routes back with that reason.
- **Interrupted.** The same run makes two more tool rounds (steps 43–50) and is then interrupted from the keyboard. Step 51 is a terminal record with status **error** and token totals of 67,586 input and 10,380 output; step 52, written after it, records the exception `KeyboardInterrupt`. The interrupt handler writes both records. The last event from the agent’s work is the tool result at step 50. The trace does not establish what the process was doing between that event and the interrupt.

A `SIGKILL` at the same moment would leave the file ending at step 50, a tool result, without terminal or exception records. No code would run to raise `NoTerminalStatus`. The first 50 records would be identical. **The two files would differ only in the records written after the work stopped.**

Neither recorded run ends with `budget_exhausted`. The following description comes from the code. The configuration accepts smaller budgets, such as ten calls. A run that reached this budget would record a routing event with decision `budget_exhausted`, `by="runtime"`, and the call count and budget, followed by the terminal record. No model call would occur between them because admission precedes the call. The gathered evidence would remain in `observations`.

3.7 Reading

LangChain [9], the sections on conditional edges, `END` and the recursion limit; and Cornell CS 2112 [5], the argument for loop correctness through its Termination step. §3.3 of Yao et al. [20] describes the repeated-action failure mode that stall checks detect.

For reference while writing HW1: the terminal statuses and trace events of `trace.py`, which define the vocabulary used in Table 5.

Looking ahead. After routing selects a location, the runtime schedules its computation. The next section examines scheduling and tool execution: what a tool contract specifies beyond the model’s schema, how the dispatcher processes a request, and which component refuses operations the run is not permitted to perform.

4 Scheduling, tools, and execution authority

4.1 Scheduling: when an update becomes visible

Routing selects the next location; scheduling determines when its computation runs. The scheduler also determines which record the computation reads and when later computations can read its update.

Definition 4.1 (Superstep). “Pregel computations consist of a sequence of iterations, called supersteps.” By design, the model “doesn’t expose any mechanism for detecting order of execution within a superstep, and all communication is from superstep S to superstep $S+1$ ” [15, §1].

A runtime using this model divides each superstep into three phases: i) *planning* selects computations whose inputs were written in the previous superstep; ii) *execution* runs those computations, whose writes remain invisible to other computations during this phase; iii) *update* applies the writes through the reducers of Section 2.3. The next superstep plans from the updated record. In the notation of Section 2.1, execution computes u_t , and the update phase computes $\text{merge}(s_t, u_t)$.

This sequence guarantees update order. The runtime applies a node’s writes after the node returns, then plans its successor. **A later node never reads a partly applied update.** Nodes in the same superstep cannot see one another’s writes. The runtime combines their writes in the update phase. If two nodes write the same field, that field needs a reducer that combines their values; otherwise, the runtime must reject the competing writes.

This guarantee applies only to the record. A tool can change a file during execution, before the superstep ends. If the node then fails, the file remains changed, but the record lacks the update that would describe the change. **Superstep visibility orders record updates. It does not order external effects.** The tool boundary must control those effects.

4.2 The tool contract

The model receives a tool’s name, description, and argument schema. The runtime holds its implementation, enforces its operating constraints, and handles invalid arguments. The description alone does not provide these controls.

Definition 4.2 (Tool contract). A *tool contract* specifies five parts of a tool: i) its name and purpose, ii) typed arguments and their constraints, iii) the typed success result and error representation, iv) its side-effect class, and v) execution constraints, such as the directories it may access. The definition supplied to the model covers the first two parts. The implementation and dispatcher determine the other three, which the model cannot establish from the description.

A method specification assigns obligations to the caller and the implementer (16, Specification structure).

Definition 4.3 (Precondition). “The precondition is an obligation on the client (the caller of the method). It is a condition over the state in which the method is invoked.”

Definition 4.4 (Postcondition). “The postcondition is an obligation on the implementer of the method.” “In general, the postcondition is a condition on the state of the program after the method is invoked, assuming the precondition was true before.”

For a tool, the precondition covers typed arguments and their constraints, including execution constraints on the request. For example, a path may have to resolve inside the workspace. The postcondition covers the result, error representation, and side effects. The precondition defines when the tool’s promises apply; the postcondition defines those promises.

A method specification may make no promises when the caller violates a precondition. An agent’s tool contract needs stricter rejection rules because the model can supply invalid arguments during normal operation. The contract must state which violations the runtime detects, how it reports refusal, and that a refused request has no side effects. **A tool contract must define how the runtime handles violated preconditions.**

Our three tools illustrate this contract. `read_file` takes a path and an optional line range and returns text; `search` takes a regular expression and an optional glob and returns matching lines with their locations; `edit` replaces one exact occurrence of a string in a file. Only `edit` writes. All three take the workspace root as their first parameter. Before opening a file, `read_file` and `edit` resolve its path through that root. `search` matches its glob against the root without resolving paths (Section 4.6).

The function signature determines which arguments the dispatcher accepts, so it also determines where configuration can be supplied. Tools read the run’s configuration from the module. They do not accept it as a parameter. The dispatcher binds model arguments against the signature, so **a parameter the dispatcher has not already supplied can receive a value from the model.** Adding a `config` parameter would let the model set it. The dispatcher itself supplies the workspace-root parameter, which is absent from the model’s schema. If a request names that parameter anyway, binding fails because the dispatcher has already supplied the argument.

4.3 The dispatch cycle

The dispatch cycle has five operations, each of which can end the cycle. Figure 7 follows one request through them.

- **Resolve the name.** The dispatcher refuses requests for unregistered tools and lists the available tools in its response. It builds the registry from the tool enumeration. A tool absent from that registry cannot run, regardless of the definitions supplied to the model.

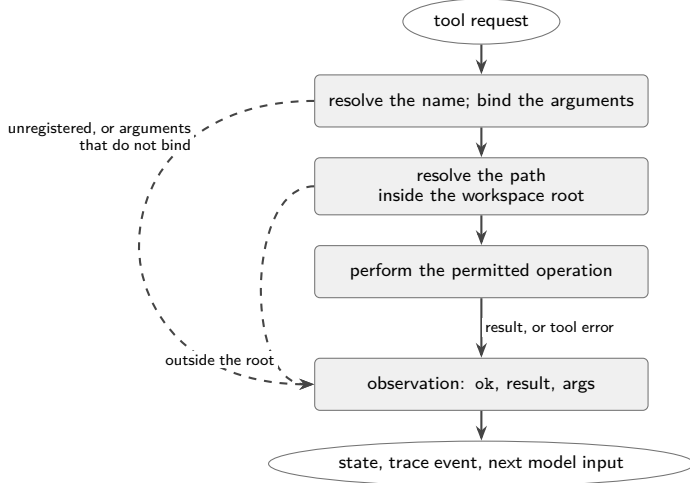


Figure 7: The logical stages of one dispatch. Ellipses show the incoming request and the records that receive the observation; grey boxes show runtime work. Solid arrows follow successful checks. Dashed arrows show refusals, which also produce observations. A policy check may run inside the tool, as in our implementation, if it precedes the protected effect.

- **Bind the arguments.** The dispatcher binds the model’s arguments against the function’s signature before calling it. A wrong argument name therefore produces a readable refusal rather than a `TypeError` from inside the tool. Binding checks names and arity. It does not check types. The dispatcher coerces unambiguous type mismatches, avoiding an extra model call to correct, for example, a string supplied where an integer was required.
- **Check the policy.** Before opening a file, the tool resolves its path and refuses it if it lies outside the workspace root. This check is the first operation in `read_file` and `edit`. The dispatcher converts any exception from this check into a refusal.
- **Perform the permitted operation.** The function runs with the bound arguments and the dispatcher’s workspace root.
- **Observe.** Success and failure alike become an `Observation` carrying the tool, the arguments, an `ok` flag and a string result, and the dispatcher writes a `tool_result` event either way.

A refusal returns an observation to the model, allowing the reactive loop to continue. An exception ends the run. If `edit` refuses an ambiguous target, the model can use the observation to retry with more context. Raising an exception would end the run at that first imprecise edit. Exceptions are reserved for unrecoverable failures, such as an unwritable trace.

The tools node dispatches multiple calls from one response sequentially within one superstep. Each call receives an observation, including failed calls. Omitting a response to any call makes the next model message malformed.

4.4 Evidence of a request and its execution

One tool call appears in six places, each of which supports different claims. Table 6 follows the first `read_file` call of the planning agent’s accepted run, `bc284c6b`, through these records and processing stages.

Step 4, between the response and the request, is the routing event: decision `tools`, `by="model"`, because the reply carried a tool call (Section 3.1). It records the selected control location. It does not describe tool execution.

The model response and the `tool_result` event support different claims. A tool call in a model response establishes what the model requested. It does not establish a file-system operation. If the model produces both a request and a plausible result in

Artifact	What it contains	What it establishes
Model input	The tool definitions and the conversation so far. The trace does not record it.	Which operations and descriptions were supplied to the model.
Model response, and the <code>model_call</code> event (step 3)	A tool call with an identifier, a name and arguments. The event keeps only the name, <code>read_file</code> .	That the model requested an operation. It does not establish execution.
<code>tool_request</code> event (step 5)	The tool name and the arguments as the dispatcher received them.	That the request reached the dispatcher and began the cycle.
Dispatcher	Registry lookup, argument binding, path resolution. No event of its own; a failed check would appear as the reason in the result event.	That the request satisfied the contract, or why it failed a check.
<code>tool_result</code> event (step 6)	<code>ok=true</code> , 1,665 characters, which is the length of the file.	That the function ran and returned, and how much text came back.
State and next input	One record appended to <code>observations</code> , one tool message appended to <code>messages</code> .	What the next model call, at step 7, was allowed to see.

Table 6: Records and processing stages for one `read_file` call on `testcode/BenchmarkTest00283.py`, from run `bc284c6b`, steps 3 to 7. Step numbers count trace events rather than supersteps. The message history retains the provider’s raw request; the trace records the request received by the dispatcher.

a transcript, neither establishes execution. In this example, the runtime’s `tool_result` event has `ok=true`, establishing that the function ran and returned. A result event can also report a refusal before function entry. The successful event does not establish the returned text’s contents, because it stores only the result’s length.

4.5 Execution authority

The model proposes actions; the runtime executes or refuses them. Each responsibility belongs to an identifiable component on the request’s execution path. Establishing that the checks cover *every* execution path requires a separate argument.

Definition 4.5 (Complete mediation). “To attempt to secure this resource, server needs to check all accesses to the resource. ‘Complete mediation.’ ” “Server will put a ‘guard’ in place to mediate every request for this particular resource. Only way to access the resource is to use the guard.” [8, §5]

For an agent, a resource is anything a tool can affect, including files, processes, or the network. The request is a tool call, and the guard consists of the checks performed before its effect. Complete mediation requires coverage of all execution routes. A correct check on one tested route says nothing about a route that bypasses it. We must therefore enumerate the routes to each protected operation. Coverage is most easily lost through an overlooked route, such as a new tool or an argument that identifies a file without being named a path. Complete mediation ensures that the runtime consults the policy. It does not establish that the policy is correct.

The term “control” involves two separate distinctions.

- **Decision timing and ownership.** Some choices are fixed when we write the program and others are computed during execution. Timing alone does not identify who controls the choice. A routing function and a limit check both run during execution, but both apply rules the programmer wrote.
- **Proposal and execution.** A requested action, a claimed result, or a declaration that the work is finished takes effect only through the runtime’s execution path. A model’s statement that it ran the tests does not establish that tests executed.

Guarantee	Where it is enforced
Only registered tools run	The registry built from the tool enumeration, consulted before anything is called.
Arguments bind to the tool’s signature	Signature binding and coercion in the dispatcher, before the call.
No file outside the workspace is written	<code>workspace.resolve</code> , called by <code>edit</code> , the one tool that writes, on its path argument.
No file outside the workspace is read	Not enforced. <code>read_file</code> resolves its path, but the glob of <code>search</code> never reaches <code>workspace.resolve</code> .
The pinned fixture is not modified	Every tool works on a disposable copy of the fixture, and the diff compares that copy with the pinned fixture.
An exhausted budget prevents another model call	The check each model node makes before it issues its call.
A plan is revised at most once per run	The tools node, which records a stall for revision only while <code>replanned_at</code> is unset, and the routing function after tools, which reads that record (Table 4).

Table 7: Desired guarantees and the components that enforce them. The implementation does not enforce one of these guarantees. Prompt instructions enforce none of them.

`workspace.resolve` fully resolves both the root and the candidate path before comparing them. Resolution collapses `..` segments and follows symlinks, preventing either from escaping the root. A string-prefix comparison would not provide the same check: `/tmp/ws` is also a prefix of the unrelated directory `/tmp/ws-evil`. Joining an absolute path to the root yields the absolute path itself. The check refuses it if its resolved destination lies outside the root.

4.6 The demo: scheduling and request refusal

Our planning agent runs on LangGraph’s runtime, which organizes execution “into multiple steps, following the Pregel Algorithm/Bulk Synchronous Parallel model”, and whose execution phase is specified so that “channel updates are invisible to actors until the next step” [11]. Our graph has no parallel branches, so each superstep runs one node. In run `bc284c6b`, steps 3 and 4 belong to the controller’s superstep: first the model call, then the routing event. The routing function reads the record after the controller’s reply has been merged. Steps 5 and 6 belong to the next superstep, when the tools node reads that reply and dispatches its call. The dispatcher writes both trace events during execution. The observation enters `observations` only after the node returns, at the update boundary.

No recorded run contains a refused request. We therefore construct this demonstration by sending requests directly to the dispatcher, outside a graph, using a fresh workspace for the worked alert. We report the dispatcher’s responses. The first

request calls `read_file` on `testcode/BenchmarkTest00283.py`. The dispatcher permits it and returns the same 1,665 characters as step 6 of the recorded run. The second request changes only the argument.

- **The request.** `{"name": "read_file", "args": {"path": "../etc/passwd"}}`, which no model produced.
- **The refusing check.** `workspace.resolve`, the first operation of `read_file`, raises `PathEscapeException`, and the dispatcher catches it.
- **The record.** An observation with `ok=false` whose result is the refusal, `refused: '../etc/passwd' resolves outside the workspace root`, and a `tool_result` event carrying the same reason.
- **Evidence that no file was opened.** The trace holds a `tool_request` event and a failed `tool_result` event with a reason and no character count. That shows the request was refused. The code establishes that no file was opened: the check precedes the first file operation. The trace does not record file operations and cannot establish this on its own.

The two requests use the same tool and dispatcher. Their arguments determine which execution branch the runtime selects. This identifies the enforcing component described in Section 4.5.

The same exercise exposes a route that bypasses the check. The dispatcher permits a `search` whose glob reaches outside the workspace. With the glob `../*-withheld/testcode/*.py` it returns lines from the benchmark cases the workspace withholds in a sibling directory. With enough `../` segments, it returns the matching line of `/etc/passwd`. Both results have `ok=true`. **The path check guards `read_file` and `edit`; it does not guard `search`, whose glob never passes through `workspace.resolve`.** Our boundary against writing outside the workspace is complete because `edit` is the only tool that writes. The implementation does not completely mediate reads. The repair is to resolve every file `search` matches through `workspace.resolve` before reading it.

4.7 Reading

§1 and §2 of Malewicz et al. [15], for supersteps and the rule that communication crosses superstep boundaries in one direction only; the section “Specification structure” of MIT 6.031 [16], for preconditions and postconditions; and §5 and §6 of LaCurts [8], the guard model and the ways complete mediation is bypassed.

For reference while writing HW1: LangChain [11] for the plan, execution and update phases as our runtime implements them; Anthropic [3] for the exact request and result representation a provider emits and expects; and `dispatch.py` and `workspace.py` for the checks between a request and its result.

Looking ahead. In the execution model described here, a routing function selects a node and the scheduler runs it in the next superstep. The next section keeps the same machine and changes this scheduling mechanism: a computation emits an event, which the runtime dispatches to a registered handler. We then examine which component invokes the handler and when. The tool contract and dispatcher remain the same. The new scheduling mechanism requires corresponding evidence that each transition occurred once.

5 Event-triggered realization

5.1 Events, handlers and inversion of control

A graph and an event loop can run the same machine with different mechanisms for invoking application code. In the graph, a routing function selects the next node, and the scheduler runs it in the next superstep (Section 4.1). In an event loop, a computation emits an event, and the runtime invokes the code registered to handle it. The following four definitions describe

this arrangement. Their wording comes from one source [4, Event Handling; Understanding the Event Loop].

Definition 5.1 (Event). “An event is any occurrence that could necessitate a change in the program’s state.”

Definition 5.2 (Handler). Handlers are “pieces of code that understand how to respond to a particular event”; for events to reach them, they “must register with the event loop.”

Definition 5.3 (Event loop). “An event loop is a piece of code that repeatedly checks whether a new event has occurred and dispatches any detected events to code that can handle them.”

Definition 5.4 (Inversion of control). “Inversion of control is used to describe the paradigm where we write custom subroutines but allow an external entity, such as a framework, to control when these subroutines are executed.”

For an agent, an event represents either an occurrence, such as *the plan is ready* or *the checks failed*, or a request, such as *run these tools*. Both are values. A tool request event records a request to a handler. It does not establish that the handler ran. The same distinction applies to a model’s tool request in Section 4.4: requesting a file read does not establish that the read occurred.

Inversion of control separates two responsibilities. Application code defines what each handler does with its event, which fields of the record it writes, and which event it emits on completion. The framework determines which handler runs, when it runs, and the order in which it processes pending events. **Application code defines the response to an event; the framework controls its invocation.** The graph realization makes a similar division: we write node bodies and routing functions, and the scheduler runs them. The two realizations express the choice of the next computation in different places. A graph uses a routing function on an edge. An event handler chooses the type of event to emit, and the registrations map that type to a handler.

The registrations determine which event types have handlers. If the loop does not check its registrations, it can remove an event from the queue even though no handler accepts its type. Nothing then runs, although no component has decided to stop. Whether the framework reports this as an error depends on its implementation. We treat it as a defect rather than a terminal outcome.

5.2 An event-triggered version of the machine

Changing the realization changes when the runtime takes a transition. The set of transitions stays the same. A state-machine diagram does not specify when a reaction occurs. In an event-triggered model, a reaction occurs when an input is present; in a time-triggered model, it occurs on the tick of an external clock [18, slides 23–25]. The planning agent’s machine (Figure 1) retains its locations, guards and set actions under either choice. Our graph realization uses no clock: each superstep begins when the preceding one finishes. The event realization reacts when an event is present and is therefore event-triggered under this definition.

The choice can still make a transition unreachable. A guard requiring every triggering input to be *absent* cannot fire when reactions require a triggering input to be present. Each transition of the planning agent becomes enabled when the preceding computation completes, so each can wait for a completion event. A transition such as *no reply arrived within thirty seconds* needs a timer to emit an event when the interval ends. Without the timer, the runtime

never takes this transition even though it appears in the diagram.

The event realization requires a larger configuration. In the graph realization, a configuration contains a control location and a valuation of the variables, (q, v) (Definition 1.4). In the event realization it is a triple

$$(E, \mathcal{H}, v),$$

where E is the sequence of pending events, $\mathcal{H}$ is the set of running handlers, and v is the record. The runtime maintains E and $\mathcal{H}$. The record retains the steps’ data and contains neither the queue nor the set of running handlers. In a serial realization, at most one event is in flight. During active execution, either one event is pending and no handler runs, or one handler runs and no event is pending. We can then recover the location q from the handler registered for the pending event or from the running handler. A realization that permits two handlers to run at once can have configurations with no single location, and the machine in Figure 1 no longer describes such configurations. Our demo’s validation step is such a case (Section 5.4).

Four rules preserve the machine’s behaviour when an event loop realizes it.

- **Apply once, then enable.** Merge a handler’s contribution into the record exactly once before making its emitted event available for dispatch. The graph’s superstep model provides this ordering (Definition 4.1). An event loop provides it only if the handler writes before it emits and the runtime does not deliver the same event twice. If a handler emits first, its successor can read the record before the contribution appears. Delivering an event twice appends the same observations twice.
- **Distinguish idle from success.** An empty queue with no running handler means that nothing is scheduled. It does not establish how the run ended or whether termination was intentional. Only a terminal event carrying a status establishes termination, as a terminal record does in the trace (Section 3.5).
- **Stop dispatch at termination.** After dispatching the terminal event, the runtime must start no further handler, even if an event remains pending. In a serial realization, no event can be pending at that point. A realization with concurrent handlers must discard or refuse the remaining events.
- **Separate triggering events from log events.** A triggering event is an input to the dispatcher that causes a handler to run. A log event records an occurrence in the trace and triggers no computation. An event that requests tools triggers the tools handler; a trace record of the same request documents that the handler was invoked, and triggers nothing.

The third and fourth rules both apply at termination. The terminal event ends dispatch, and a terminal record in the trace records its status. If the process fails before writing that record, dispatch can end without a terminal record. The interpretations in Table 5 still apply.

5.3 Two frameworks

Two frameworks provide event dispatch through different APIs.

- **LlamaIndex Workflows.** Dispatch is by event type. A step is a method decorated with `@step`, and “A step receives an event, does some work, and returns another event. That returned event triggers the next step whose type annotation accepts it” [13]. Events are user-defined Pydantic classes, and the framework supplies two of its own: `StartEvent`, whose fields are the keyword arguments passed to `run()`, and `StopEvent`, whose return ends the workflow with the value passed as its `result`. The step signatures declare the event types each step consumes and produces. Before a run, the framework validates the event graph, checking, among other conditions, that it contains a start and a stop event and that every produced event has a consumer. A store on the workflow’s `Context` holds per-run state.

Steps read and write it through `ctx.store`, using `edit_state()` for updates that must not interleave with another step's updates [14]. Our demo is written in this framework (Section 5.4).

- **CrewAI Flows.** Dispatch uses method completion and route labels. A method decorated with `@start()` is an entry point. A method decorated with `@listen(m)` runs when method `m` completes and receives its output. A method decorated with `@router(m)` runs after `m` and returns a string label. Methods decorated with `@listen("label")` run when the router returns that label. The operators `or_` and `and_` combine triggers: a listener runs when any or all of several methods have produced output, respectively. The flow's `state` attribute holds shared state as a dictionary or a Pydantic model. There is no stop event: "the final output is determined by the last method that completes" [6]. Translated to CrewAI, each of our event types would become a router label, and the outcome σ would have to be written to `state` by our code, because the end of the flow does not carry it.

This difference affects how we distinguish idle from success. LlamaIndex has an explicit terminal event. Its pre-run validation checks the event graph declared by step signatures for start and stop events and missing event connections. These checks do not establish that a particular run will reach `StopEvent`. CrewAI defines no terminal event. Our reading of its documentation is that a flow ends when no method remains to run, corresponding to an empty queue. The outcome must therefore be in the record before the last method completes.

Neither framework establishes that an event realization preserves the machine's behaviour. Comparing it with the graph realization requires a controlled design. We compare the realizations after each computation completes and its contribution is applied. These logical boundaries define the points of comparison; internal handler steps and wall-clock ordering do not. We script the model's replies and control the tool results so that both realizations receive the same inputs in the same order. A faithful translation preserves four properties at each boundary: i) the observations appended to the record, ii) the effects on the workspace, iii) the results of the acceptance checks, and iv) the terminal status and its meaning. The comparison is designed to isolate the scheduling mechanism: a graph superstep or an event dispatch.

Agreement under scripted inputs supports a limited conclusion: the two realizations implement the same transition rules for the scripted replies. It does not establish equivalent behaviour with a live model. Model replies depend on the messages each realization constructs, so changing how a handler formats messages changes the input distribution even when the transition rules stay the same. The comparison also excludes concurrency.

5.4 The demo: the planning agent as a LlamaIndex workflow

Our event-driven agent, `cyberbird/event_driven`, runs the planning agent's machine as a LlamaIndex workflow. It keeps the machine's locations, guards and limits: a budget of forty model calls per alert, checked before each call; a stall after three identical results; and one plan revision. The run's data sit in a store on the workflow's context. A `RunState` holds the run's bookkeeping, and one `AlertState` per alert holds that alert's plan, messages, observations, calls and usage. Two of its steps are still stubs: `submit` runs no mechanical checks yet, and each validator accepts after a fixed delay without calling a model. No run of this agent has been recorded.

The six rows below the rule in Table 8 are the machine's locations. No step names another: the wiring is the set of step signatures, and the framework dispatches each event to the step whose signature accepts its type. The guards that the graph evaluated in routing

functions now sit inside the steps and choose which event to return.

- **Admission.** `plan` and `controller` first reserve one model call for the alert. When the alert has spent its budget, the step returns `AlertDone` with status `budget_exhausted` and makes no call.
- **Acting.** The controller returns `ToolsRequested` when the model's reply contains tool calls, and `Submitted` when it contains none.
- **Revising.** `tools` returns `Observed` while the work moves. On the first stall it returns `Stalled`, which the plan step accepts; that step writes a new plan and moves `replanned_at`, the handoff and window of Section 2.4. A second stall returns `AlertDone` with status `no_progress`.

The event realization also does three things the graph agent does not. First, one run handles many alerts. `scan` runs Bandit once and emits one `AlertFound` per alert, and `gate` releases one alert at a time: it queues each alert it receives and starts the next one only after the current alert's `AlertDone` arrives. Every event about an alert carries its `alert_id`, which correlates the event with that alert's entry in the store.

Second, validation fans out and back in. `submit` sends one `ValidateRequested` per validation lens (correctness, behaviour and scope) with `ctx.send_event`, because a step can return only one event. The validate step runs with up to three workers, so the three validators run at the same time. This is the configuration of Section 5.2 with more than one running handler and no single location. `judge` holds each `Verdict` until all `V` have arrived, in any order, and accepts the alert only when every validator accepted. It collects verdicts without keying them by alert, which is safe only because the gate admits one alert at a time. **The gate serializes alerts; it does not serialize validators.**

Third, the run has two kinds of ending. An alert ends with an `AlertDone` carrying one of four statuses: `accepted`, `rejected`, `no_progress` or `budget_exhausted`. That event returns to the gate, which records the status. The run ends when the gate has a result for every alert the scan found, and it returns a `StopEvent` carrying the results. A scan that finds no alert ends the run at once with an empty result.

The code keeps the four rules of Section 5.2 in these forms.

- **Apply once, then enable.** Every write goes through `ctx.store.edit_state()`, which locks the store, and each step writes its data before it emits the event that lets another step read them. The budget check reserves its call under the same lock, so two steps admitted at the same moment cannot both take the last call.
- **Distinguish idle from success.** Only the gate's `StopEvent` ends the run. Before the first run, the framework also checks that every event a step emits has a step that accepts it, which is why `submit` lists `ValidateRequested` in its return annotation although it sends those events rather than returning them.
- **Stop dispatch at termination.** The gate emits `StopEvent` only when the count of results equals the count of alerts, so no alert is still in progress when the run ends.
- **Separate triggering events from log events.** This agent writes no trace. Its events trigger steps, and no event records one. Our graph agents write a trace alongside their events (Section 3.5); the event-driven agent does not yet have one.

5.5 Reading

Cornell CS 2110 [4], the sections Event Handling and Understanding the Event Loop, for events, handlers, the event loop and inversion of control; and Seshia [18], slides 23–25, on event-triggered and time-triggered reactions.

For reference while writing HW1: LlamaIndex [13], the sections

Events the step accepts	Step	Events the step emits	Decided by
RunRequested	start	ScanRequested	runtime
ScanRequested	scan	one AlertFound per alert, or StopEvent	runtime
AlertFound, AlertDone	gate	AlertStarted, one alert at a time, or StopEvent	runtime
AlertStarted, Stalled	plan	PlanCreated, AlertDone	model
PlanCreated, Observed, ChecksFailed	controller	ToolsRequested, Submitted, AlertDone	model
ToolsRequested	tools	Observed, Stalled, AlertDone	runtime
Submitted	submit	ChecksFailed, or $V \times \text{ValidateRequested}$	runtime
ValidateRequested	validate	Verdict	model
$V \times \text{Verdict}$	judge	AlertDone	runtime

Table 8: The steps of the event-driven agent, from the wiring table in its `workflow.py`. Each step’s signature names the events it accepts, and its return annotation names the events it emits. The rows below the rule are the machine of Figure 1; the rows above it are new. V is the number of validators, three by default.

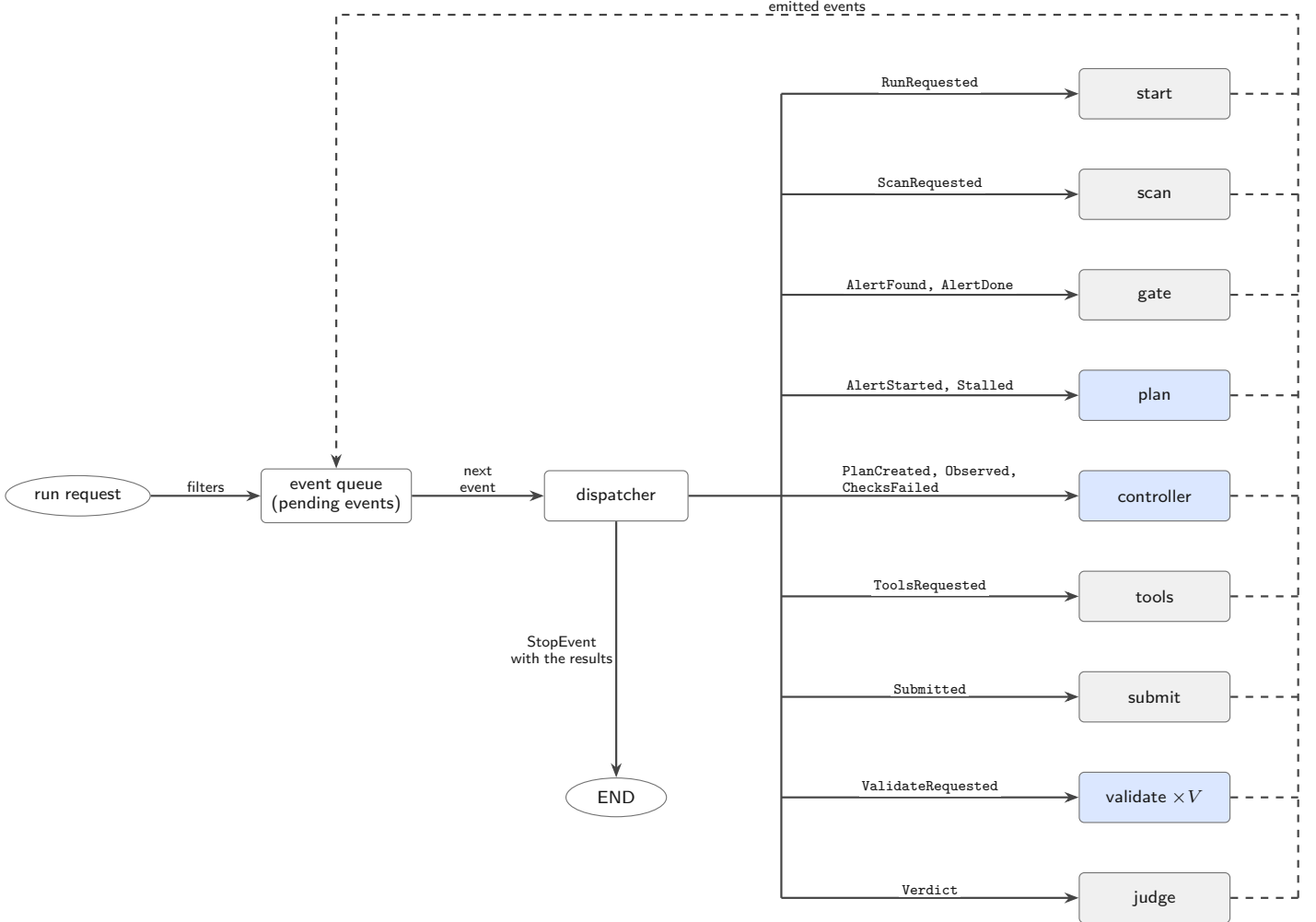


Figure 8: The event-driven agent of Table 8. White boxes are runtime components. Steps use the colours of Figure 1: blue steps call a model (the validators will, once they are no longer stubs), and grey steps run tools, checks or bookkeeping. The ellipses are the run request that starts the run and END. Each solid arrow from the dispatcher names the event types its step accepts. Every event a step emits returns to the queue along the dashed path. V validators run at the same time.

on entry points, exit points and validation, with LlamaIndex [14] for the per-run store; and CrewAI [6], the sections on `@start()`, `@listen()`, retrieving the final output, conditional logic and the router.

Looking ahead. In an event loop, a registered handler runs when its event arrives, and a handler can emit any number of events. The loop itself does not decide whether the work is permitted or assess its cost. The next section distinguishes admission, which decides whether a model or tool call may start,

from accounting, which records its cost after execution. It also considers how to measure a realization’s cost under the controlled conditions used here to compare behaviour.

6 Budgets and controlled measurement

6.1 Admission and accounting

A runtime needs two mechanisms to limit spending. Before a model or tool call, it must decide whether the call may start. After execution, it must record the cost. The first mechanism uses

information available before the call; the second uses the usage reported afterwards.

Definition 6.1 (Admission). Admission decides whether a unit of work may start. It checks the run’s existing commitments against a limit, then permits or refuses the work. Refused work never runs and incurs no charge.

Definition 6.2 (Accounting). Accounting records the cost of a unit of work after execution. It adds the incurred charge to a running total, using the same units as the limit. Admission reads this total.

Admission determines whether work may start; accounting records its cost. Each depends on the other. If accounting omits a call, the admission check can permit work that the limit would otherwise exclude. If no admission check reads the total, accounting merely records an overrun after it occurs. **Accounting maintains the usage total; admission uses it to enforce the limit.**

Accounting reads the charge reported with each reply. Admission must decide before that reply exists. It knows the input to be sent but does not yet know the output. This difference makes a call-count limit easier to enforce than a token limit: each call counts as one, while its token cost becomes known only after execution.

Our planning agent uses call-count admission. It checks a budget of forty calls before each model call and records the call’s input and output tokens. The token admission and reservation mechanism developed below is a proposed design. Our code does not implement it.

Call counts measure usage coarsely. In our two recorded runs, a single call’s input ranges from 571 to 13,550 tokens because each controller call resends the growing conversation. A budget of forty calls therefore bounds the number of replies. The resulting bound on token cost depends on the largest possible call.

6.2 A budget invariant

Admission and accounting need an explicit budget unit. The unit determines which cost the budget limits. We consider three choices.

- **Output tokens.** Count only tokens generated by the model. A per-call output limit bounds each call’s contribution in advance, simplifying admission. This budget excludes input tokens, whose count grows as calls resend the conversation.
- **All processed tokens.** Count input and output tokens together to track the provider’s work. Admission requires bounds on both the input and output of each call.
- **Money.** Weight tokens by their prices. Token kinds can have different prices, as with cache reads and fresh input. Accounting must therefore retain separate counts until it computes the charge. Price changes alter how much work the budget permits.

The following argument applies to all three units. The unit determines how we calculate the charge γ of a call.

We express the budget as an invariant over three quantities. B is the budget. S is the settled spend, the sum of recorded charges for completed calls. R is the sum of outstanding reservations, the amounts set aside for admitted calls that have not yet been settled. The invariant is

$$S + R \leq B.$$

A *reservation* ρ is the amount set aside for a call before it starts. It must be a justified upper bound on that call’s cost. *Settlement* replaces the reservation with the recorded charge γ after the call completes.

The required property concerns the charges themselves: the total charge for all calls started by the run, including unfinished calls, must never exceed B . As with the call counter in [Section 2.5](#), the runtime can compare only recorded values. To connect the record’s invariant to the charges, we require

$$\forall i. \gamma_i \leq \rho_i,$$

where i ranges over admitted calls, ρ_i is the reservation for call i , and γ_i is its charge on completion. This obligation constrains how admission computes ρ . It is not a state predicate: the final charge γ_i of an outstanding call is not yet known to the runtime. The input and output bounds below justify this obligation. If the obligation holds, $S + R$ bounds the total charge of all calls started, and the invariant implies the required property. We use the invariant method ([Definition 2.3](#)) to prove that the inequality holds initially and that each transition preserves it.

- **Initially.** No call has been admitted or settled, so $S = 0$ and $R = 0$. The invariant therefore holds for any $B \geq 0$.
- **Admit.** If $S + R + \rho \leq B$, admission reserves ρ for the call by setting R to $R + \rho$. The guard ensures that the invariant holds after this step. Otherwise, admission refuses the call and leaves the record unchanged.
- **Settle.** The call completes with charge γ . Settlement sets S to $S + \gamma$ and R to $R - \rho$, changing their sum by $\gamma - \rho$. Because $\gamma \leq \rho$, the sum cannot grow, and $S + R \leq B$ still holds. This step also releases the unused amount $\rho - \gamma$ to the unreserved budget.
- **Release.** If an admitted call is never issued, for example because the run ends first, the runtime removes its reservation and reduces R by ρ . The call incurred no charge, so the invariant holds.
- **Every other transition.** Leave S and R unchanged. In particular, no transition decreases S . Settled spend has the run lifetime defined in [Section 2.4](#). Resetting it to zero would make the budget available again while leaving the charges incurred.

Concurrent calls require one further condition: the guard check and reservation must form a single step. Suppose two model calls are admitted in the same superstep ([Section 4.1](#)). If both check $S + R + \rho \leq B$ before either adds its reservation to R , both can pass while their combined cost exceeds B .

The proof depends on establishing $\gamma \leq \rho$. For a budget measured in all processed tokens, ρ is the sum of an input bound and an output bound. Each needs justification.

- **Input.** Establish an upper bound on the call’s input tokens. The provider’s counting endpoint returns an estimate before the call is sent: “The token count is an estimate. In some cases, the actual number of input tokens used when creating a message might differ by a small amount” [2]. Comparing past estimates with recorded charges can inform a margin. A strict guarantee, however, requires a justified upper bound on the estimation error. Alternatively, admission can use a limit the request cannot exceed, such as the model’s context window.
- **Output.** Use an enforced upper bound on output tokens. The provider enforces the per-call output limit sent with the request and stops generation when that limit is reached [1]. This bound holds regardless of the content the model intended to generate.

An estimate alone cannot justify the invariant. If admission computes ρ from an estimate and the call costs more, settlement increases the sum by $\gamma - \rho > 0$. The record contains no flag identifying the step at which $S + R$ exceeded B . **This reservation-based guarantee requires an upper bound on each call’s charge.** Estimates support a forecast of spending. The claim that spending cannot exceed B requires the bounds above.

The runtime must also handle unused reservations and uncertain charges. Releasing unused reservations is a liveness requirement: an unreleased reservation keeps R high and causes admission

to refuse work that would otherwise fit. Leaving an unused reservation outstanding does not violate the spending invariant. An uncertain charge requires a different response. If a timeout or crash prevents receipt of a reply, the provider may still have charged for the call. Releasing its reservation could omit that charge from the record. The reservation must remain outstanding until the runtime reconciles the charge with the provider’s usage records. Only then can it settle the call or release the reservation.

6.3 Exhaustion and truncation

A budget and a per-call limit stop work at different points. The budget prevents new work from starting. The per-call limit stops generation within an admitted call. A run can reach either limit, with a different result in each case.

- **Exhaustion.** When admission refuses the next call, that call never starts. The run ends with a declared status, and a terminal event in the trace records the reason (Table 5). Exhaustion leaves calls already in flight running until they complete and are settled.
- **Truncation.** When a call reaches its output limit, the provider stops generation and reports the reason in the response. It returns a successful response containing the content generated before the limit.

Truncation matters most for tool requests. The provider’s documentation describes the problem: “If Claude’s response is cut off because it hit the `max_tokens` limit, and the truncated response contains an incomplete tool use block, you’ll need to retry the request with a higher `max_tokens` value to get the full tool use” [1]. A truncated request can still resolve to a known tool and bind to its signature. For example, an edit request can have valid argument names but incomplete replacement text. The dispatch checks in Section 4.3 test names, arity and policy; they do not check completeness. **The stop reason can flag a potentially incomplete tool request even when it passes every dispatch check.** The runtime must therefore read the stop reason before dispatch. It must refuse an incomplete request before any effect occurs and return an observation on which the model can act.

Recovery from truncation charges both the original call and the retry. The truncated call ran and incurred a charge, so accounting settles it normally. A retry with a larger output limit is a new call. It requires a separate admission check and a reservation sized for the larger limit.

Admission also applies to calls after the main work. Retries, plan revisions and final assessments generated by a model each require an admitted model call. If the controller uses the entire budget, the run has no budget left for assessment and ends exhausted. In our planning agent, the validator uses the same admission check as the other two model nodes. If a patch passes its checks after the fortieth call, the run ends `budget_exhausted` without a verdict. A design that requires a final assessment reserves part of the budget for it in advance. Other roles are then admitted against B minus that allocation. Our code does not implement this reservation.

Accounting must cover every role and attempt. Counting only controller calls would leave the planner’s and validator’s spending unlimited. Keeping a budget per attempt would reset it after each failed attempt. Settled spend must have the run lifetime (Section 2.4), as the call counter does across the proposed attempt edge in Section 3.3. In our planning agent, all three model nodes update a shared call counter and usage total. Each model-call event in the trace identifies its role, so the trace can reconstruct call counts by role.

6.4 Controlled measurement

Comparing the cost of two realizations requires a common budget and a controlled design. The scripted comparison in Section 5.3

tests whether two realizations implement the same transition rules for the scripted replies. It does not measure behaviour with a live model, whose replies depend on the messages each realization constructs. A live comparison measures the model and harness together: “When we evaluate ‘an agent,’ we’re evaluating the harness and the model working together” [7]. Its unit is the trial.

Definition 6.3 (Trial). “Each attempt at a task is a trial. Because model outputs vary between runs, we run multiple trials to produce more consistent results.”

A comparison varies the realization, holds other conditions fixed, and repeats each configuration over several trials. It holds seven conditions fixed.

- **Tasks.** The same alerts, in the same fixture, for every realization.
- **Initial environment.** Each trial starts from a fresh copy of the same workspace. This prevents access to files, caches or history from earlier trials only if tools cannot read outside the workspace. Our tools permit such access through the `glob` argument of `search`, as Section 4.6 shows.
- **Model configuration.** The same model, version, sampling parameters and per-call output limit.
- **Prompts.** The same system prompts and task messages for each role.
- **Tools.** The same tool set, with the same signatures, descriptions and limits.
- **Acceptance rules.** The same checks and the same validator, so that *accepted* means the same thing in both arms.
- **Budget policy.** The same unit, B , and admission rule. A realization that makes more calls reaches the limit sooner. Different budget policies could make a policy effect appear to be an effect of the realization.

For each realization, we report the distribution of terminal outcomes across trials, together with i) usage by token kind, ii) calls by role, iii) latency, and iv) the exhaustion rate. Acceptance rates alone cannot distinguish rejection from budget exhaustion. Usage alone cannot distinguish low-cost completion from an early stop.

A fixed seed does not establish equivalent trajectories. It fixes how the sampler draws from a distribution for a given input. It does not fix that input. If two realizations format messages differently, the model receives different inputs from the first call. A difference in one tool result also changes all later messages. Whether identical requests produce identical outputs depends in part on how the provider serves them, which the seed does not control. Our planning agent fixes seed 0 and temperature 0. These settings fix the sampler’s configuration without determining the trajectory. Repeated trials make it possible to measure differences between realizations.

Such a measurement supports conclusions about the tested realizations, model, prompts, tasks and budget. It does not establish which realization performs better with another model, which may respond differently to the same messages, or under another budget, which changes where runs stop. A small number of trials may also be insufficient to separate a real difference from variation between runs.

6.5 Reading

revisit MIT 6.042 [17], §2 (p. 5), for the proof method used in the reservation invariant; and read Grace et al. [7], the sections The structure of an evaluation, How to think about non-determinism in evaluations for agents, and Design the eval harness and graders, on trials, the harness, and isolated environments.

For reference while writing HW1: Anthropic [2], on what the

counting endpoint counts and its warning that the result is an estimate; and Anthropic [1], the sections on `max_tokens`, incomplete tool use blocks, and handling truncated responses.

Looking ahead. In this account of an agent, a named component makes each decision, and a named artifact records it. A routing function selects the next location, and a routing event records the choice. The dispatcher permits or refuses a tool request, and an observation records the result. Admission permits a call, and accounting records its cost. The runtime sets the terminal status, and the trace records it. HW1 asks you to describe your agent in the same terms. For each decision, identify the responsible component and the artifact that provides evidence of the decision. If either is missing, a reviewer cannot check what occurred.

References

- [1] Anthropic. Stop reasons and fallback. Claude Platform documentation, <https://platform.claude.com/docs/en/build-with-claude/handling-stop-reasons>, 2026. Accessed 2026-09-24. HW1 reference material, not assigned reading.
- [2] Anthropic. Token counting. Claude Platform documentation, <https://platform.claude.com/docs/en/build-with-claude/token-counting>, 2026. Accessed 2026-09-24. HW1 reference material, not assigned reading.
- [3] Anthropic. Tool use with claude. <https://platform.claude.com/docs/en/agents-and-tools/tool-use/overview>, 2026. Accessed 2026-09-18. HW1 reference material, not assigned reading.
- [4] Cornell CS 2110. 25. Event-Driven Programming. Lecture notes, CS 2110 Object-Oriented Programming and Data Structures, Cornell University, <https://courses.cs.cornell.edu/cs2110/2025fa/lectures/lec25/>, 2025. Fall 2025. Accessed 2026-09-24.
- [5] Cornell CS 2112. Loop invariants. Lecture notes, CS 2112, Cornell University, <https://www.cs.cornell.edu/courses/cs2112/2022fa/lectures/loopinv/>, 2022. Fall 2022. Accessed 2026-09-24.
- [6] CrewAI. Flows. <https://docs.crewai.com/v1.15.22/en/concepts/flows>, 2026. Documentation version 1.15.22. Accessed 2026-09-24. HW1 reference material, not assigned reading.
- [7] Mikaela Grace, Jeremy Hadfield, Rodrigo Olivares, and Jiri De Jonghe. Demystifying evals for AI agents. Anthropic Engineering, <https://www.anthropic.com/engineering/demystifying-evals-for-ai-agents>, 2026. January 9, 2026. Accessed 2026-09-24.
- [8] Katrina LaCours. Lecture 20 outline. Lecture outline, 6.033 Computer System Engineering, Massachusetts Institute of Technology, MIT OpenCourseWare, <https://ocw.mit.edu/courses/6-033-computer-system-engineering-spring-2018/pages/week-11/lecture-20-outline/>, 2018. Spring 2018. Accessed 2026-09-24.
- [9] LangChain. LangGraph graph API. <https://docs.langchain.com/oss/python/langgraph/graph-api>, 2026. HW1 setup material, not assigned reading.
- [10] LangChain. Persistence. <https://docs.langchain.com/oss/python/langgraph/persistence>, 2026. Accessed 2026-09-18. HW1 reference material, not assigned reading.
- [11] LangChain. LangGraph runtime. <https://docs.langchain.com/oss/python/langgraph/pregel>, 2026. Accessed 2026-09-24. HW1 reference material, not assigned reading.
- [12] Jukka Lehtosalo. PEP 589: TypedDict, type hints for dictionaries with a fixed set of keys. <https://peps.python.org/pep-0589/>, 2019. Accessed 2026-09-18. Final; incorporated into the typing module.
- [13] LlamaIndex. Agent workflows: Introduction. <https://developers.llamaindex.ai/python/llamaagents/workflows/>, 2026. Accessed 2026-09-24. HW1 reference material, not assigned reading.
- [14] LlamaIndex. Agent workflows: Managing state. https://developers.llamaindex.ai/python/llamaagents/workflows/managing_state/, 2026. Accessed 2026-09-24. HW1 reference material, not assigned reading.
- [15] Grzegorz Malewicz, Matthew H. Austern, Aart J. C. Bik, James C. Dehnert, Ian Horn, Naty Leiser, and Grzegorz Czajkowski. Pregel: A system for large-scale graph processing. In *Proceedings of the 2010 ACM SIGMOD International Conference on Management of Data (SIGMOD)*, pages 135–146, 2010. Copy used: <https://15799.courses.cs.cmu.edu/fall2013/static/papers/p135-malewicz.pdf>. Accessed 2026-09-24.
- [16] MIT 6.031. Reading 6: Specifications. Course reading, 6.031 Software Construction, Massachusetts Institute of Technology, <https://web.mit.edu/6.031/www/sp21/classes/06-specifications/>, 2021. Spring 2021. Accessed 2026-09-24.
- [17] MIT 6.042. Lecture 16: State machines and invariants. Lecture notes, 6.042 Mathematics for Computer Science, Massachusetts Institute of Technology, <https://courses.csail.mit.edu/6.042/past-devel/archive/spring03/handouts/lectures/lec16.pdf>, 2003. April 10, 2003. Accessed 2026-09-24.
- [18] Sanjit A. Seshia. Discrete dynamics: State machines. Lecture slides, EECS 149/249A Introduction to Embedded Systems, UC Berkeley, <https://ptolemy.berkeley.edu/projects/chess/eeecs149/lectures/DiscreteDynamics.pdf>, 2015. Slides © 2008–2015 E. A. Lee, A. L. Sangiovanni-Vincentelli, S. A. Seshia. Accessed 2026-09-24.
- [19] Sanjit A. Seshia. Extended finite state machines, timed automata, hybrid automata. Lecture slides, EECS 149/249A Introduction to Embedded Systems, UC Berkeley, <https://ptolemy.berkeley.edu/projects/chess/eeecs149/lectures/ExtendedAndTimedAutomata.pdf>, 2015. Slides © 2008–2015 E. A. Lee, A. L. Sangiovanni-Vincentelli, S. A. Seshia. Accessed 2026-09-24.
- [20] Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. ReAct: Synergizing reasoning and acting in language models. In *International Conference on Learning Representations (ICLR)*, 2023. ICLR version, arXiv:2210.03629v3.

A Conceptual exercises

For every question, select all correct options. Each question has one or more correct options. Explain why your selected options are correct and why the others are incorrect. The answers and explanations follow in [Appendix B](#).

A.1 How an agent moves through a task

Related material: [Section 1](#).

1. **Same step, different situation.** An agent returns to its decision step after reading a file. It is about to run the same decision code as before. Which statements explain how its next action can change?
 - (A) The saved file result can change what the decision step chooses.
 - (B) Returning to the same step restores the saved information to its earlier value.
 - (C) The agent can be at the same step while remembering a different conversation and call count.
 - (D) A different next action requires adding a new step to the workflow.
2. **Replies and rules.** Two runs reach the decision step with identical saved information. The model replies differently in the two runs. The program has a fixed rule for choosing the next step from the reply. Which statement best describes the result?
 - (A) The fixed rule forces the model to give the same reply in both runs.
 - (B) The runs may take different routes, while the rule still gives one route for each saved state and reply.
 - (C) Different routes show that the rule must choose randomly.
 - (D) The rule assigns a probability to each reply.
3. **Repeating a run.** A team wants to reproduce an agent run exactly. They have the same workflow code and starting information. What else would make the same sequence of decisions possible?
 - (A) The same model replies, while tool results may vary without consequence.
 - (B) The same tool results, while model replies may vary without consequence.
 - (C) A list of all routes the workflow allows, without the results that selected them.
 - (D) The same relevant model replies and tool results in the same order.
4. **A small workflow with a long memory.** A workflow has five kinds of step. It keeps every tool result in a list with no fixed length limit. Which statements follow?
 - (A) The workflow can still have just five kinds of step.
 - (B) The number of distinct saved situations can have no finite bound.
 - (C) Every new tool result requires a new kind of step.
 - (D) The small number of steps makes it possible to list every reachable saved situation in a finite table.
5. **What needs remembering.** Two visits to the controller have the same latest tool result. In one run, two model calls have been completed; in the other, 40 have been completed, reaching the call limit. Which statements explain what the agent must remember?
 - (A) The latest tool result alone is enough to decide whether another model call is allowed.
 - (B) The completed-call total must remain available even when the latest tool result is the same.
 - (C) The program must retain every earlier model reply solely to enforce the numeric call limit.
 - (D) Saved information should preserve past facts that can change the next decision, even if it does not preserve every past detail.
6. **What a tool request changes.** A model reply asks the agent to run a tool. The program checks for that request, sends it to the tool, saves the reply, and counts the completed model call. Which statements are accurate?
 - (A) The model reply is information that arrived during this step.
 - (B) The check for a tool request changes the saved call count.
 - (C) Sending the request to the tool is an action the step asks the outside world to perform.
 - (D) Saving the reply and adding one to the call count change what later steps can read.
7. **Possible paths.** A diagram shows a recorded route in blue and other routes in grey. The grey routes depend on model replies or check results that did not occur in the recorded run. Which interpretation is justified?
 - (A) Each grey route was taken once, but its events were omitted from the log.
 - (B) A grey route is more likely than a blue route because it has fewer steps.
 - (C) The blue route is one actual path among several paths the rules permit.
 - (D) The grey routes become impossible once the blue route is recorded.
8. **What a log can prove.** A log says a file-reading tool succeeded and reports the length of its result. It omits the result text. Some moves between steps also have no log event. Which conclusions are justified?
 - (A) The exact file text can be reconstructed from the reported length.
 - (B) The log can establish that the recorded tool call succeeded.
 - (C) Every saved value after the tool call can be read directly from the log.
 - (D) A missing move may be inferred from nearby events and the workflow rules, with that inference identified as such.
9. **Plan versus workflow.** A planner changes its list of subgoals after seeing repeated tool results. The agent's workflow still has the same named steps. Which statements follow?
 - (A) The new plan can affect later actions when the controller reads it.
 - (B) Each new subgoal becomes a new named step in the workflow.
 - (C) The controller may respond differently even though the workflow wiring did not change.
 - (D) The program chooses the next step by directly testing the name of the current subgoal.
10. **A second stall.** The agent has already revised its plan once. Since that revision, three new tool results are identical and show no successful edit. Which statements match the workflow?
 - (A) The earlier results that triggered revision must be deleted before the new pattern can be noticed.
 - (B) The new repeated results can cause a stop for lack of progress.
 - (C) The plan is revised again each time three results match.
 - (D) The agent can distinguish the new results from the old ones using the saved position where revision occurred.

A.2 What an agent remembers between steps

Related material: [Section 2](#).

11. **A small update.** The controller detects that the call limit has been reached and returns an update that changes only the run's status. What happens to the conversation and observations already saved?
 - (A) They remain in the working record unless another update changes them.
 - (B) They are removed because the controller did not return them.
 - (C) The controller must copy them into its update to keep them.
 - (D) The status can change without changing either history.
12. **A limit outside the prompt.** The program refuses to make a model call after 40 completed calls. The model is never told the number of calls used. Which statement best explains how the limit can still work?
 - (A) The model will infer the count from the length of the conversation.
 - (B) The model's reply can cancel a call before the call is made.
 - (C) The tool history automatically prevents the 41st model call.
 - (D) The program keeps a count and checks it before every model call.
13. **Surviving a process stop.** A run stops unexpectedly. Its event log and edited files remain on disk, but the agent saved no checkpoint of its working information or pending step. Which statements follow?
 - (A) A new process can resume at the exact pending step from the event log alone.
 - (B) The saved files may show effects of work that the agent already performed.
 - (C) The surviving log is useful for studying the run but does not provide everything needed for resumption.
 - (D) The in-memory conversation and counters necessarily survive because the files do.
14. **Keep, replace, or add.** The agent needs all tool results to detect repetition, only the latest plan for current work, and a total of completed model calls. Which update choices fit those needs?
 - (A) Keep adding new tool results to the saved result list.
 - (B) Replace the old plan when a revised plan becomes current.
 - (C) Add one to the call total for each completed model call.
 - (D) Replace the call total with one after each model call.
15. **An empty list is not a reset.** The agent keeps tool results by appending each update to a saved list. A step returns an empty list, intending to clear the history. Which statements are accurate?
 - (A) The saved history stays as it was after appending the empty list.
 - (B) The empty update clears the history because it has no elements.
 - (C) A direct replacement with an empty list would produce the intended reset.
 - (D) The history disappears from the list of declared fields.
16. **A message meant for one step.** A tool step records why progress has stalled so the planner can revise the plan. The planner reads that reason. What should happen next?
 - (A) Leave the reason active until the revised plan succeeds.
 - (B) Clear the reason after the planner uses it.
 - (C) Clear the reason and erase the full tool-result history at the same time.
 - (D) Keep the reason in a growing list and add an empty list after the planner reads it.
17. **New evidence after revision.** Three repeated tool results cause a plan revision. The agent keeps the full result history because other steps need it. How can it judge whether work stalls again after revision?
 - (A) Save the position in the result list where the new plan began.
 - (B) Check only results added after that saved position.
 - (C) Treat the old three results as the first three results after revision.
 - (D) Discard the full history whenever the plan changes.
18. **Retries and a whole-run budget.** Imagine the agent is extended to retry a failed patch. It clears the failed candidate and check results before the retry. Which choices keep a 40-call limit across all attempts?
 - (A) Keep the call total when the candidate and checks are cleared.
 - (B) Reset the call total at each retry because the candidate is new.
 - (C) Check the retained total before each model call, including calls in the retry.
 - (D) Tell the model about the limit in its prompt instead of keeping a total.
19. **Counting every call.** The program stores a completed-call count and refuses a call when the count reaches 40. Which practices are needed to establish that completed calls stay within the limit?
 - (A) Increase the stored count once for each completed call.
 - (B) Check the stored count only in the controller, even though the planner and validator also call the model.
 - (C) Leave the count unchanged on steps that make no model call.
 - (D) Let a plan revision reset the count if it remains below 40 afterward.
20. **Declarations and checks.** An agent uses a Python type annotation to list the names and expected types of its saved fields. An editor checks the annotation, but the program adds no run-time validator. Steps may return updates for only some fields. Which statements follow?
 - (A) The annotation can help the editor detect a step that refers to an undeclared field.
 - (B) The annotation alone rejects every wrongly typed value returned while the agent runs.
 - (C) The agent should still start with values for fields its first steps will read.
 - (D) Because updates can be partial, every field in the starting information may safely be omitted.

A.3 Choosing the next action and knowing when to stop

Related material: [Section 3](#).

21. **What the next step knows.** A step reads a search result but saves only the fact that the search finished. The next-action check needs to decide whether the result mentions the flagged line. Which statements about what this check can use are true?
 - (A) The check can use the original result because the earlier step saw it.
 - (B) The check can tell that the search finished, but not whether it mentioned the flagged line.
 - (C) It could use whether the flagged line was mentioned if the earlier step had saved that detail.
 - (D) The check can treat the fact that the search finished as evidence that the flagged line appeared in its result.

22. **Two checks agree.** A workflow tests two possible reasons to continue. For one saved state, both tests say yes. The program tries them in a fixed order and takes the first match. What follows?
- (A) Both next steps run, since both tests matched.
 - (B) The first matching test determines the next step.
 - (C) The fallback step runs because the two tests conflict.
 - (D) The model's wording determines which test takes priority.
23. **A request and a suggestion.** A model reply says, 'Submit the patch now,' and also requests a tool. The run is still active. The programmer's rule sends replies with tool requests to the tools step. Which statements hold?
- (A) The tool request is a value that affects the next-action check.
 - (B) The sentence about submitting overrides the program's rule.
 - (C) The model has changed the rule for future replies.
 - (D) The programmer's rule determines what the request means for the next step.
24. **Different runs, same workflow.** The same compiled workflow handles two alerts. One run revisits the controller many times; the other submits quickly. Which explanations fit?
- (A) Every revisit requires the workflow to be compiled again.
 - (B) Different saved results can make the same next-action rule choose different routes.
 - (C) Each controller visit is a separate moment in that run, even when it visits the same step.
 - (D) Compiling the workflow fixes the exact sequence of steps for every run.
25. **Learning from a failed check.** A proposed patch fails a mechanical check. The workflow sends the reasons back to the controller so it can try again. Which choices support that repair?
- (A) Keep the failed proposal and the check's reasons available to the controller.
 - (B) Erase the conversation so the controller cannot see the failed proposal.
 - (C) Allow the controller to make a new proposal that replaces the stored patch.
 - (D) Keep the overall call limit in force while repairs continue.
26. **Changing a stalled plan.** Three repeated observations show that the current plan is stalled. The workflow allows one plan revision. Which actions make the revision meaningful?
- (A) Send the planner back with no reason for the stall.
 - (B) Give the planner the stall reason and start measuring later observations under the revised plan.
 - (C) Discard every earlier observation before the planner can revise.
 - (D) Stop the run if the revised plan later stalls again under its own observations.
27. **Counting new attempts.** A proposed extension would start a fresh attempt after a final reviewer rejects a patch. It has limits on both attempts and total model calls. What must remain across attempts to make those limits work?
- (A) The rejected patch must remain the active patch for the next attempt.
 - (B) The old plan must remain active because changing it changes the workflow.
 - (C) The attempt count and spent-call count must survive the reset.
 - (D) A written reflection can replace an attempt count.
28. **A limit with a gap.** A run has a limit on model calls. An added tool retry can repeat without calling the model, and a model call can wait indefinitely for a reply. Which concerns remain?
- (A) The tool retry needs its own bound or another reason it must stop.
 - (B) The call limit alone makes every individual operation finish.
 - (C) A termination argument also needs operations to return or time out.
 - (D) A successful-patch check ensures the run eventually finds a successful patch.
29. **A narrow stall detector.** Every search returns a different irrelevant result, so no three results are identical. The run still makes no useful progress. What follows?
- (A) A check for identical results can miss this kind of stalled work.
 - (B) Changing results show that the search is getting closer to a solution.
 - (C) Because the stall check does not fire, the run must eventually succeed.
 - (D) A check before each model call can still stop new calls at the budget.
30. **What an ending shows.** One trace ends with an accepted status. Another ends after a tool result and has no recorded ending. A third ends with an error status followed by an exception record. Which conclusions are supported?
- (A) The accepted status proves the patch truly satisfies the task.
 - (B) The accepted status shows the run ended and its acceptance procedure passed.
 - (C) The trace without an ending does not establish whether the run terminated.
 - (D) The error status identifies the kind of ending, and the following record identifies the exception.

A.4 Tool requests, execution, and access

Related material: [Section 4](#).

31. **When updates appear.** The runtime holds proposed updates until every worker in a batch has finished. Two workers read the saved state at the start of the same batch, and one finishes first. Which statements about saved state follow?
- (A) It reads the state available when the batch began.
 - (B) It sees the first worker's update as soon as that worker finishes.
 - (C) Workers in a later batch can read the combined update after it is applied.
 - (D) If both update one field, the runtime can use whichever finished last without a combination rule.
32. **A file changed before a crash.** A tool edits a file, then its worker crashes before the worker's update to the saved run state is applied. What is possible?
- (A) The file edit is automatically undone with the missing state update.
 - (B) The file has changed even though the saved state does not describe the change.
 - (C) The next worker must see the crashed worker's partial state update.
 - (D) The file change becomes visible only when the worker's saved-state update is applied.

33. **Instructions and enforced limits.** A tool description tells the model to stay inside a workspace, but a search request can still read a sibling folder. Which conclusions follow?
- (A) The instruction in the description cannot itself prevent the tool from reading outside files.
 - (B) The description makes outside reads impossible when the model follows it most of the time.
 - (C) The model can decide whether a particular outside path is permitted by explaining its intent.
 - (D) The actual files reached by search must be checked before they are read.
34. **Who can change a setting.** A new tool takes a run setting as an argument. The dispatcher does not supply that argument, so a model request can fill it. The setting should be controlled by the application. Which change achieves that?
- (A) Ask the model in the tool description to leave the setting unchanged.
 - (B) Give the setting a default value while leaving it available as an argument.
 - (C) Keep the setting outside arguments the model can supply, or have trusted runtime code supply it.
 - (D) Accept the model's value whenever it gives a convincing reason.
35. **A well-formed unsafe request.** One request uses the wrong argument name. Another uses the correct name but points outside the allowed workspace. A third uses a clear string where an integer is expected. What follows?
- (A) Checking argument names also proves a path is permitted.
 - (B) The wrong name can be refused before the tool runs, while the path needs an access check before opening the file.
 - (C) If the path argument binds to the tool, the file must be opened.
 - (D) The dispatcher can correct an unambiguous type mismatch without treating it as permission to cross the workspace boundary.
36. **A useful refusal.** An edit request matches two places in a file, and the tool refuses to choose one. How should a continuing run handle this?
- (A) Return a failed result with the reason to the model.
 - (B) Edit the first match so the model gets some progress.
 - (C) Make no edit for the refused request.
 - (D) Allow a later request with enough surrounding context to identify one occurrence.
37. **Evidence that a tool ran.** A trace shows a model asked to read a file, the request reached the dispatcher, and a successful result records the number of characters returned. What does this establish?
- (A) The model's request by itself proves the file was read.
 - (B) The successful result supports that the reading function ran and returned.
 - (C) The trace does not reveal the exact returned text from its character count.
 - (D) The request record alone shows that all access checks passed.
38. **A new way to read files.** A team has tested that its file-reading tool blocks paths outside the workspace. Later it adds another tool that can read files without using that check. What must the team consider?
- (A) Every route that can read a protected file needs a check before the read.
 - (B) The first tool's test results cover the new tool because both read files.
 - (C) A prompt instruction to remain in the workspace closes the new route.
 - (D) A shared gateway helps only if all file-reading routes actually pass through it.
39. **Where a path really leads.** A workspace is at `/tmp/ws`. A request names a path containing `..`, a symbolic link, or an absolute path. A sibling folder named `/tmp/ws-copy` also exists. Which checks make sense?
- (A) Accept any path whose text begins with `/tmp/ws`.
 - (B) Resolve the workspace and candidate to their real destinations before comparing them.
 - (C) Treat a path that starts inside the workspace as potentially outside after resolution.
 - (D) Treat an absolute path as inside the workspace once it has been joined to the workspace path.
40. **One unchecked search route.** A search tool accepts a pattern reaching a sibling folder and returns its lines. The editing tool checks its target path before writing. What can be concluded?
- (A) The write boundary has failed because search changed the matching files.
 - (B) The read boundary has a gap through search.
 - (C) Checking each matched file's resolved destination before reading would close this search route.
 - (D) A successful containment test for the other reading tool does not prove search is contained.

A.5 Events and queued work

Related material: [Section 5](#).

41. **A request is waiting.** A planning assistant puts "run these file checks" in a work queue. The worker has not picked it up. What does the queued item tell us?
- (A) The checks have run because the request is visible in the queue.
 - (B) The worker may run the checks when the runtime delivers that item.
 - (C) The queue has already changed the files being checked.
 - (D) If the request is later delivered, successful execution needs evidence beyond the queued request itself.
42. **Who runs the worker?** A team writes a function to handle a completed plan and registers it with an event loop. Who controls the response, and who decides when to call the function?
- (A) The team decides what the function does when a plan arrives.
 - (B) The function must directly call every worker that might follow it.
 - (C) The runtime decides when to invoke the registered function.
 - (D) The runtime invents the function's rules for updating the plan.

43. **A message with no reader.** A worker finishes planning and puts “plan ready” in the queue, but no worker is registered to receive that kind of item. The runtime removes it and the queue becomes empty. What follows?
- (A) The task succeeded because the queue is empty.
 - (B) A planned next step may have been lost.
 - (C) No part of the application necessarily chose to stop.
 - (D) The runtime can recover the missing worker from the plan text.
44. **Quiet is not success.** A run has no queued work and no worker running. It also has no recorded outcome. Which conclusion is justified?
- (A) The run succeeded because every queue is now empty.
 - (B) The last worker must have decided to stop.
 - (C) The runtime cannot tell from quietness alone how the run ended.
 - (D) The run necessarily failed its file checks.
45. **Save before notifying.** A tools worker gets a file result and wants to notify the controller that results are ready. What ordering protects the next decision?
- (A) Notify the controller first; it can infer the file result from the words “results ready.”
 - (B) Save the file result before the notification can be delivered.
 - (C) It is safe to save after notifying because the queue is empty before the notification.
 - (D) Apply the result once so the controller does not read a duplicate observation.
46. **One message, two deliveries.** A worker has already added one tool result to the saved history. The runtime accidentally delivers the same completion message again. What risk does this create?
- (A) The second delivery proves the tool ran a second time.
 - (B) The first saved result is automatically removed.
 - (C) The controller may act twice on one result or append that result twice.
 - (D) The duplicate makes an explicit stopping outcome unnecessary.
47. **No reply arrives.** A model call should time out after thirty seconds. This event loop starts work only when an item arrives. What makes the timeout path reachable?
- (A) Waiting silently for thirty seconds automatically invokes the timeout worker.
 - (B) A timer can put a timeout item in the queue when the interval ends.
 - (C) A note saying “still waiting” in the log is enough to invoke the worker.
 - (D) The timeout needs some source of a new item if the model never replies.
48. **Work after stopping.** A run declares its outcome, but a “run tools” item remains in the queue. How should the runtime treat it?
- (A) It must start the tools worker before recording the outcome.
 - (B) The pending item alone changes the declared outcome to success.
 - (C) It may ignore the outcome if the queue still contains work.
 - (D) It must prevent that pending item from starting a worker.
49. **One worker or several.** A simple version allows only one pending item or one running worker at a time. A new version allows two workers to run together. What changes in how we describe its progress?
- (A) With one worker at a time, the pending item or running worker identifies the current step.
 - (B) With two workers running, one step name always describes everything that is happening.
 - (C) Allowing two workers means the saved information no longer matters.
 - (D) With two workers running, the workflow may need a description that includes both active steps.
50. **Comparing two ways to run it.** A team runs the same workflow once by following graph arrows and once by delivering queued items. Both versions run one worker at a time, and the team supplies the same prepared model replies and tool results in the same order. What can comparing saved information after each completed step show?
- (A) Matching observations, file effects, checks, and outcomes support agreement for those prepared inputs.
 - (B) The two versions must take the same amount of clock time.
 - (C) Agreement does not show what a live model would say if the two versions build different messages.
 - (D) The comparison does not cover two workers acting at once.
- ## A.6 Budgets and spending
- Related material: [Section 6](#).
51. **Permission and the bill.** A model provider reports the token charge only after a call finishes. The run has a spending limit. Which jobs must the runtime do?
- (A) Decide before the call whether it is allowed to start.
 - (B) Wait for the bill before deciding whether that same call may start.
 - (C) Add the actual charge to its spending record after the call.
 - (D) Assume every model call has the same token cost.
52. **Spent and set aside.** A run has a limit of 10 units. It has spent 4 and set aside 4 for a call still in progress. A proposed new call could cost up to 3. What follows?
- (A) It may start because only 4 units have been billed so far.
 - (B) It should be refused for now because the spent amount plus both possible calls would exceed 10.
 - (C) The unfinished call can be ignored until its final bill arrives.
 - (D) If the unfinished call later costs only 2, the runtime can reconsider the new call.
53. **Forecast or safe limit?** A provider’s input-token counter gives an estimate, while the request sets an enforced maximum for output tokens. The team wants a strict guarantee that charges never exceed its token budget. Which conclusions follow?
- (A) The input estimate alone guarantees the call cannot cost more than the amount set aside.
 - (B) The amount set aside must cover a justified upper bound on the whole call’s cost.
 - (C) An enforced output limit gives a bound for that part of the call.
 - (D) Past estimates that were accurate once prove all later estimates will be upper bounds.
54. **Two calls at once.** There is room for either of two proposed calls, but not both. Two workers check the available amount at the same moment. What prevents both from starting?
- (A) Let each worker check first, then set money aside later.
 - (B) Start both and correct the record after their bills arrive.
 - (C) Make checking the remaining amount and setting money aside one indivisible action.
 - (D) Give both calls the same output limit.

55. **A call times out.** A model call times out before the runtime receives a reply. The provider might still bill for work already done. What should happen to the money set aside for that call?
- (A) Release it at once because a missing reply means the provider did no work.
 - (B) Record a zero charge and give the full amount to the next call.
 - (C) Keep it set aside until the provider's usage record shows the actual charge.
 - (D) Treat the timeout as a failed attempt and reset all spending for the run.
56. **What does the limit count?** The controller resends a growing conversation on every model call. A team wants the limit to reflect that extra input work. Which choices meet or explain that goal?
- (A) Counting only generated output captures the growing input as well.
 - (B) Counting all processed tokens includes both the sent conversation and generated reply.
 - (C) A money limit may need separate counts for token kinds with different prices.
 - (D) Counting calls fixes the token cost of each call.
57. **An unfinished tool action.** A model reply hits its output limit. It names a valid editing tool and valid arguments, but the replacement text ends partway through. What should the runtime do?
- (A) Run the edit because the tool name and argument names are valid.
 - (B) Inspect why generation stopped before allowing the edit to affect files.
 - (C) Discard the model call's charge because its proposed edit was unusable.
 - (D) Count the original call and check the budget again before a retry with a larger output limit.
58. **Leave room to judge.** A run needs one final model call to assess a patch after file checks pass. The controller can otherwise use all available calls first. Which policy protects the final assessment?
- (A) Hold back enough of the run's allowance for the assessment before admitting other calls.
 - (B) Let the controller use the whole allowance and treat assessment as free.
 - (C) Reset the spending total when file checks pass.
 - (D) Skip the starting check for the final call because it comes after the main work.
59. **Trying again.** A future version starts a fresh attempt after a failed patch. It clears the old candidate and checks. Which choices keep the spending limit meaningful across attempts?
- (A) Keep the amount already spent across the whole run.
 - (B) Reset spending along with the candidate so each attempt has a full new allowance.
 - (C) Count planner, controller, and final-assessment calls, including retries.
 - (D) Count only calls that lead to an accepted patch.
60. **Comparing live runs.** A team compares a graph version and a queue-driven version with a live model. Which choices support a useful comparison, and what can it show?
- (A) Run the same tasks with the same model settings, prompts, tools, checks, and budget in both versions.
 - (B) Use one trial per version because the same random seed makes their replies identical.
 - (C) Repeat trials from fresh workspaces, restrict tools to those workspaces, and report outcomes alongside usage and exhaustion rates.
 - (D) Limit the conclusion to the tested model, tasks, prompts, and budget.

B Answers and explanations

The numbers match the questions in [Appendix A](#). Each answer identifies all correct options and explains every choice. Question numbers link back to the exercises.

B.1 How an agent moves through a task

Related material: [Section 1](#).

1. **Correct: A, C.**
 - (A) **Correct.** The step reads saved information, so a new observation can lead it down a different route.
 - (B) **Incorrect.** Revisiting a step does not undo updates made by earlier steps.
 - (C) **Correct.** The step name identifies the work to do; the saved values identify the particular situation.
 - (D) **Incorrect.** The existing step can choose among its existing routes using changed information.
2. **Correct: B.**
 - (A) **Incorrect.** The program's routing rule does not produce or control the external reply.
 - (B) **Correct.** Different replies can satisfy different checks; each specific reply still has a definite route.
 - (C) **Incorrect.** The difference can come entirely from different replies, without any random choice in the rule.
 - (D) **Incorrect.** A routing rule selects an action after a reply arrives; it does not by itself model reply likelihood.
3. **Correct: D.**
 - (A) **Incorrect.** Tool results can enter saved information and change later decisions.
 - (B) **Incorrect.** Model replies can request different tools or request submission.
 - (C) **Incorrect.** Allowed routes show possibilities, but they do not identify which one occurred.
 - (D) **Correct.** With the same starting information and the same inputs at each step, the fixed rules select the same routes.
4. **Correct: A, B.**
 - (A) **Correct.** The list changes the saved information, not the number of named steps.
 - (B) **Correct.** Lists of arbitrarily many results can distinguish arbitrarily many situations.
 - (C) **Incorrect.** The existing tool and decision steps can process another result.
 - (D) **Incorrect.** An unbounded result list may make that table infinite.
5. **Correct: B, D.**
 - (A) **Incorrect.** The same result occurs in both runs, but the call limit permits another call in only one of them.
 - (B) **Correct.** The program needs that total to distinguish a run below the limit from one at the limit.
 - (C) **Incorrect.** An accurate running total is enough for this limit; individual replies may be retained for other purposes.
 - (D) **Correct.** The call count matters to the next admission check, while irrelevant details can be omitted from this decision.
6. **Correct: A, C, D.**
 - (A) **Correct.** The program could not derive the particular reply from its saved information alone.
 - (B) **Incorrect.** The check decides whether a route applies; the count changes through an update.
 - (C) **Correct.** This step issues a request for tool execution. Later handling may check the request before the tool runs.
 - (D) **Correct.** Those updates become part of the information retained between steps.

7. **Correct: C.**
 - (A) **Incorrect.** A possible route is not evidence that it occurred.
 - (B) **Incorrect.** The diagram assigns no probabilities, and path length does not supply one.
 - (C) **Correct.** The diagram separates the route taken from other permitted continuations.
 - (D) **Incorrect.** They remain possible for other inputs or runs, even though this run took the blue route.
8. **Correct: B, D.**
 - (A) **Incorrect.** Many different texts have the same length.
 - (B) **Correct.** Success is one of the facts the event explicitly records.
 - (C) **Incorrect.** The log omits the result text and much of the agent's saved information.
 - (D) **Correct.** The rules can explain which move connects recorded events, but that move was not logged directly.
9. **Correct: A, C.**
 - (A) **Correct.** The plan is passed to the controller, whose replies can then change.
 - (B) **Incorrect.** Subgoals are stored task data; the workflow's step names stay fixed.
 - (C) **Correct.** Its input now includes a revised plan and possibly new observations.
 - (D) **Incorrect.** In this agent, routing depends on replies and checks, not on which subgoal is current.
10. **Correct: B, D.**
 - (A) **Incorrect.** The agent keeps the full history and checks only the results since revision.
 - (B) **Correct.** After one revision, the stall route ends the run instead of revising again.
 - (C) **Incorrect.** The workflow permits one revision; a later stall has a different outcome.
 - (D) **Correct.** That position marks the start of the post-revision observation window.

B.2 What an agent remembers between steps

Related material: [Section 2](#).

11. **Correct: A, D.**
 - (A) **Correct.** A step that omits a field leaves its stored value in place.
 - (B) **Incorrect.** Steps return only fields they change; omission is not a deletion.
 - (C) **Incorrect.** The runtime retains omitted fields without asking the node to repeat them.
 - (D) **Correct.** The runtime applies the update to the named field and keeps the other stored values.
12. **Correct: D.**
 - (A) **Incorrect.** Conversation length is not a reliable count, and the model does not enforce this limit.
 - (B) **Incorrect.** The reply arrives after the call, so it cannot perform the pre-call check.
 - (C) **Incorrect.** A history records events but does not itself reject a call.
 - (D) **Correct.** The runtime can enforce the limit using saved information that is absent from the model's context.
13. **Correct: B, C.**
 - (A) **Incorrect.** The log contains selected events rather than the full working information and pending work.
 - (B) **Correct.** Workspace artifacts can survive even when the agent's in-memory information is lost.
 - (C) **Correct.** It records selected actions and results, not a complete resumable state.
 - (D) **Incorrect.** Files and in-memory information have different lifetimes.

14. **Correct: A, B, C.**

- (A) **Correct.** The repetition check needs earlier results, so replacing the list would discard needed evidence.
- (B) **Correct.** The new plan supersedes the old one for later work.
- (C) **Correct.** Each call contributes a delta to a total that lasts for the run.
- (D) **Incorrect.** That would lose the calls made before the latest one.

15. **Correct: A, C.**

- (A) **Correct.** An empty list contributes no new elements to an append operation.
- (B) **Incorrect.** Appending zero elements is different from replacing the stored list.
- (C) **Correct.** A reset must bypass the append rule and write the new list as the stored value.
- (D) **Incorrect.** An update changes a value; it does not remove the field declaration.

16. **Correct: B.**

- (A) **Incorrect.** A later route could treat the old reason as a new stall before the revised plan has been evaluated.
- (B) **Correct.** The signal has served its intended reader and should not influence a later routing decision.
- (C) **Incorrect.** The one-use signal should be cleared, but the tool-result history remains useful to later checks and validation.
- (D) **Incorrect.** Adding an empty list would leave the old reason in place, so it would not clear the active signal.

17. **Correct: A, B.**

- (A) **Correct.** That boundary separates results from before and after the revision.
- (B) **Correct.** Earlier repeated results should not count as new evidence of another stall.
- (C) **Incorrect.** They were produced under the previous plan and would cause an immediate false stall.
- (D) **Incorrect.** A saved boundary provides a fresh window while preserving history for other readers.

18. **Correct: A, C.**

- (A) **Correct.** The budget applies to the whole run, so earlier calls must still count.
- (B) **Incorrect.** This could admit more than 40 calls across several attempts.
- (C) **Correct.** Every place that can issue a call must honor the same limit.
- (D) **Incorrect.** A prompt does not supply a reliable program check before a call is issued.

19. **Correct: A, C.**

- (A) **Correct.** The stored count must match the calls actually completed for the check to be sound.
- (B) **Incorrect.** A call from another step could bypass the limit.
- (C) **Correct.** Changing it without a call would break its meaning as the completed-call total.
- (D) **Incorrect.** The reset would hide earlier calls and permit the run to exceed the whole-run limit.

20. **Correct: A, C.**

- (A) **Correct.** The declared field names give the static checker information to compare with the code.
- (B) **Incorrect.** The editor's static check does not install run-time validation for returned values.
- (C) **Correct.** A field marked optional in the declaration can still cause a missing-value problem if a running step expects it.
- (D) **Incorrect.** Partial updates leave existing values alone; they do not create values needed by an early reader.

B.3 Choosing the next action and knowing when to stop

Related material: [Section 3](#).

21. **Correct: B, C.**

- (A) **Incorrect.** The check receives saved information; seeing a result earlier does not make its discarded contents available later.
- (B) **Correct.** That is the only fact this step saved for the later check.
- (C) **Correct.** A check can use a relevant detail when an earlier step preserves it in the run's state; in this scenario, that detail was discarded.
- (D) **Incorrect.** Completion says the search ran; it does not say what the search found.

22. **Correct: B.**

- (A) **Incorrect.** A first-match rule selects one next step, even when several tests are true.
- (B) **Correct.** The program's test order resolves the overlap for this state.
- (C) **Incorrect.** The fallback is reached only when none of the earlier tests selects a step.
- (D) **Incorrect.** The priority is written into the program's test order, not inferred from the model's prose.

23. **Correct: A, D.**

- (A) **Correct.** The program examines whether the reply contains a request and uses that saved fact to choose a route.
- (B) **Incorrect.** The rule reads the request, not the persuasive wording of the reply.
- (C) **Incorrect.** A model reply supplies data to the rule; it does not rewrite the program's routing logic.
- (D) **Correct.** The rule sends an active reply containing a tool request to the tools step.

24. **Correct: B, C.**

- (A) **Incorrect.** A cycle can revisit a step within one compiled workflow.
- (B) **Correct.** The rule is fixed, while the values it reads can differ across runs.
- (C) **Correct.** A repeated location still has its own state and place in the run's sequence of decisions.
- (D) **Incorrect.** Compilation fixes possible routes; run-time values select one route at each decision.

25. **Correct: A, C, D.**

- (A) **Correct.** The next proposal can use that evidence to address the specific failure.
- (B) **Incorrect.** The repair path retains the conversation, including the failed proposal and reasons.
- (C) **Correct.** A repair needs a new candidate while preserving the evidence that explains the previous failure.
- (D) **Correct.** This repair path has no separate retry counter, so the run's call limit bounds repeated attempts.

26. **Correct: B, D.**

- (A) **Incorrect.** Without the recorded reason, the planner loses evidence of what failed and may repeat the same approach.
- (B) **Correct.** The planner can respond to the failure, and later checks then judge the new plan on its own evidence.
- (C) **Incorrect.** The workflow keeps observations; it changes the point from which the stall window is measured.
- (D) **Correct.** Only one revision is allowed, so a second stall ends the run instead of starting another revision.

27. **Correct: C.**

- (A) **Incorrect.** A fresh attempt must clear the old patch so it is not mistaken for the new candidate.
- (B) **Incorrect.** Plans are run data; replacing one does not change the workflow's structure.
- (C) **Correct.** Resetting either count would make used attempts or calls available again and could defeat their limits.
- (D) **Incorrect.** A reflection explains the failure but does not measure how many attempts have already been used.

28. **Correct: A, C.**

- (A) **Correct.** The model-call count does not change during that retry cycle.
- (B) **Incorrect.** A limit on the number of calls says nothing about the duration of one call or tool operation.
- (C) **Correct.** A decreasing count only bounds repeated cycles if each step between decreases finishes.
- (D) **Incorrect.** The check recognizes success when it occurs; it cannot force a future patch to pass.

29. **Correct: A, D.**

- (A) **Correct.** The observed failure pattern is irrelevant but varied, so it does not meet the identical-result condition.
- (B) **Incorrect.** Difference alone says nothing about relevance to the task.
- (C) **Incorrect.** A detector for one failure pattern supplies no success or termination guarantee for other patterns.
- (D) **Correct.** The budget applies to calls regardless of whether a stall pattern is detected.

30. **Correct: B, C, D.**

- (A) **Incorrect.** It shows that the acceptance procedure passed, which can still miss a defect in the patch.
- (B) **Correct.** Those are the two claims recorded by an accepting terminal event.
- (C) **Correct.** It could reflect an active run or a process stopped before recording a terminal event.
- (D) **Correct.** The terminal status and exception record carry different parts of the explanation.

B.4 Tool requests, execution, and access

Related material: [Section 4](#).

31. **Correct: A, C.**

- (A) **Correct.** Writes from this batch are applied after the workers finish, so finish order does not expose a partial update.
- (B) **Incorrect.** Updates remain invisible to other workers in the same batch until the batch's update phase.
- (C) **Correct.** The runtime applies the batch's writes before planning later work.
- (D) **Incorrect.** Competing writes need a specified way to combine them or must be rejected.

32. **Correct: B.**

- (A) **Incorrect.** The batch ordering rule governs saved state, not changes a tool already made to a file.
- (B) **Correct.** The external edit can happen during execution before the worker returns a state update.
- (C) **Incorrect.** A worker's writes become visible only when the runtime applies its completed update.
- (D) **Incorrect.** A file edit is an external effect and can occur before any update to saved run state is applied.

33. **Correct: A, D.**

- (A) **Correct.** Text shown to the model may influence its requests, but the execution path needs an access check to enforce the boundary.
- (B) **Incorrect.** A boundary must handle any request that reaches the tool, including a mistaken or hostile one.
- (C) **Incorrect.** Permission belongs to the runtime's policy, not to the model's account of why it wants a file.
- (D) **Correct.** Checking each file at the point of access closes the route that the description left open.

34. **Correct: C.**

- (A) **Incorrect.** A request can still supply the argument; an instruction does not remove that route.
- (B) **Incorrect.** A caller can override a default whenever the argument remains bindable.
- (C) **Correct.** Then the model has no open argument slot through which to replace the run setting.
- (D) **Incorrect.** A justification in the request does not transfer authority over trusted configuration.

35. **Correct: B, D.**

- (A) **Incorrect.** A correctly named path can still refer to a forbidden location.
- (B) **Correct.** Name binding and file permission are separate checks at different stages of execution.
- (C) **Incorrect.** Binding establishes that an argument fits the signature; the tool can still refuse it under the path policy.
- (D) **Correct.** Type correction handles a representation mistake; it does not change the access policy.

36. **Correct: A, C, D.**

- (A) **Correct.** The reason tells the model why the request failed and what to clarify.
- (B) **Incorrect.** The request did not identify which occurrence to change, so picking one risks an unintended edit.
- (C) **Correct.** A refusal must happen before the protected effect, leaving the file unchanged for that request.
- (D) **Correct.** A recoverable refusal lets the model refine its request and continue the run.

37. **Correct: B, C.**

- (A) **Incorrect.** A request can be refused or fail before the tool executes.
- (B) **Correct.** The result records success after dispatch rather than merely a proposed action.
- (C) **Correct.** A length reports how much text came back, not which characters it contained.
- (D) **Incorrect.** The request record precedes the checks; their outcome appears in the result.

38. **Correct: A, D.**

- (A) **Correct.** A tested check on one route leaves the new route able to bypass the boundary.
- (B) **Incorrect.** The new tool follows a different execution path and may skip the tested check.
- (C) **Incorrect.** The instruction may influence requests but does not enforce access at execution time.
- (D) **Correct.** Centralizing the check is effective when no alternate route reaches the file without it.

39. **Correct: B, C.**

- (A) **Incorrect.** That text also prefixes /tmp/ws-copy, and links or '..' can change the actual destination.
- (B) **Correct.** Resolution accounts for parent segments and symbolic links before deciding containment.
- (C) **Correct.** A link or parent segment can lead out even when the initial text appears local.
- (D) **Incorrect.** Joining an absolute path can leave it absolute; its resolved destination still needs the containment check.

40. **Correct: B, C, D.**

- (A) **Incorrect.** Search reads matching files; the editing tool is the writer and checks its target.
- (B) **Correct.** Search can return outside lines without checking each matched file against the workspace.
- (C) **Correct.** The check would refuse a match whose real location is outside the workspace before its contents are read.
- (D) **Correct.** The two tools reach files through different paths, so one passing test cannot establish the other's behavior.

B.5 Events and queued work

Related material: [Section 5](#).

41. **Correct: B, D.**

- (A) **Incorrect.** A queued request records intended work. The worker has not yet executed it.
- (B) **Correct.** Delivery to a worker registered for that kind of work is what starts the computation.
- (C) **Incorrect.** Putting an item in a queue does not perform the worker's file operations.
- (D) **Correct.** A record of execution or a successful result can show the checks ran; the queued request shows only intent. A refusal result would show that the checks did not run.

42. **Correct: A, C.**

- (A) **Correct.** Application code contains the steps that read the plan, update saved information, and choose what work comes next.
- (B) **Incorrect.** The function can put a new item in the queue; the runtime delivers it to the registered worker.
- (C) **Correct.** The loop checks pending items and calls the function registered for their kind.
- (D) **Incorrect.** Those rules come from the team's code, even though the runtime controls invocation.

43. **Correct: B, C.**

- (A) **Incorrect.** An empty queue only says that no work is scheduled; it does not record a successful outcome.
- (B) **Correct.** The completed plan was supposed to lead to another computation, but nothing can receive its message.
- (C) **Correct.** Work can cease because delivery failed, without a worker making a stopping decision.
- (D) **Incorrect.** Dispatch uses registered item kinds, and no registration exists for this one.

44. **Correct: C.**

- (A) **Incorrect.** A quiet queue can result from missing work as well as intentional completion.
- (B) **Incorrect.** The last worker could have failed to schedule its continuation.
- (C) **Correct.** A declared outcome is needed to distinguish success, a planned stop, and a lost continuation.
- (D) **Incorrect.** Queue state says nothing about the result of checks that may or may not have run.

45. **Correct: B, D.**

- (A) **Incorrect.** That message identifies the next kind of work but does not contain the full saved result.
- (B) **Correct.** The controller then reads the updated information when it starts.
- (C) **Incorrect.** The runtime may deliver the new item immediately, before the later save happens.
- (D) **Correct.** Repeating the same update could make one tool result appear twice in the saved history.

46. **Correct: C.**

- (A) **Incorrect.** Repeating a message is a delivery error and does not establish a second tool execution.
- (B) **Incorrect.** A repeated message does not undo an append to history.
- (C) **Correct.** Two deliveries can start duplicate handling of the same occurrence.
- (D) **Incorrect.** Duplicate delivery does not explain when or why the run should end.

47. **Correct: B, D.**

- (A) **Incorrect.** Silence provides no item for this loop to deliver.
- (B) **Correct.** That item gives the loop an event it can deliver to the timeout worker.
- (C) **Incorrect.** A log entry records an occurrence but is not necessarily an item submitted for dispatch.
- (D) **Correct.** Without a reply or timer event, no input arrives to start the timeout transition.

48. **Correct: D.**

- (A) **Incorrect.** The decision to end the run should prevent further work from starting.
- (B) **Incorrect.** An item requesting work does not determine the status of a run.
- (C) **Incorrect.** A declared stop is the point after which no new worker should start.
- (D) **Correct.** The terminal decision ends dispatch even if stray work remains.

49. **Correct: A, D.**

- (A) **Correct.** There is only one active continuation, so it maps to one point in the original workflow.
- (B) **Incorrect.** The workers may be at different parts of the workflow at the same time.
- (C) **Incorrect.** Both workers still read and update saved information, and their timing now matters more.
- (D) **Correct.** A single current-step label cannot represent simultaneous computations at different locations.

50. **Correct: A, C, D.**

- (A) **Correct.** These are the observable consequences of the transition rules at each comparison point.
- (B) **Incorrect.** The comparison fixes inputs and step boundaries, not scheduling overhead or timing.
- (C) **Correct.** Different messages can lead a live model to give different replies even when the step rules match.
- (D) **Correct.** The prepared comparison describes one continuation at a time, so concurrent cases require separate analysis.

B.6 Budgets and spending

Related material: [Section 6](#).

51. **Correct: A, C.**

- (A) **Correct.** A limit can prevent a new charge only if the permission check happens before work begins.
- (B) **Incorrect.** The bill does not exist yet, so it cannot be used for that call's starting decision.
- (C) **Correct.** Later starting decisions need an accurate total of charges already incurred.
- (D) **Incorrect.** The input grows with the conversation, and outputs can have different lengths.

52. **Correct: B, D.**
- (A) **Incorrect.** The 4 units already set aside must also be counted while the earlier call is unfinished.
 - (B) **Correct.** Starting it would commit up to 11 units in total: 4 spent, 4 set aside, and 3 more.
 - (C) **Incorrect.** Ignoring it would let multiple calls claim the same remaining capacity.
 - (D) **Correct.** Releasing the unused 2 units would leave enough room to set aside 3 for a later call.
53. **Correct: B, C.**
- (A) **Incorrect.** The provider warns that actual input usage can differ from the estimate.
 - (B) **Correct.** If the final charge can exceed the set-aside amount, the run can overspend when the call finishes.
 - (C) **Correct.** Generation stops at the requested maximum, so output use cannot exceed that limit.
 - (D) **Incorrect.** Past agreement can guide a forecast, but it does not establish a bound for every future request.
54. **Correct: C.**
- (A) **Incorrect.** Both may see the same available amount and each may decide that its call fits.
 - (B) **Incorrect.** The spending limit could already have been exceeded by then.
 - (C) **Correct.** The second worker then sees the first worker's commitment before deciding.
 - (D) **Incorrect.** Equal limits do not make their combined possible cost fit the remaining budget.
55. **Correct: C.**
- (A) **Incorrect.** The provider may have processed part or all of the request before the timeout.
 - (B) **Incorrect.** A zero bill is an unsupported guess while the actual charge is unknown.
 - (C) **Correct.** The amount remains committed until the runtime can record or rule out the bill.
 - (D) **Incorrect.** A failed attempt can still incur a charge, and earlier charges also remain part of the run.
56. **Correct: B, C.**
- (A) **Incorrect.** Output-only counting excludes the tokens sent into the model.
 - (B) **Correct.** Input and output token counts both contribute to this unit.
 - (C) **Correct.** Fresh input, cache reads, and output can have different prices before being combined into one charge.
 - (D) **Incorrect.** One call can resend a much longer conversation than another.
57. **Correct: B, D.**
- (A) **Incorrect.** Those checks do not show that the replacement text is complete enough to apply safely.
 - (B) **Correct.** The output-limit stop reason warns that a seemingly valid request may be cut off.
 - (C) **Incorrect.** The model call still ran and consumed provider resources.
 - (D) **Correct.** A retry is another paid call whose larger possible output needs its own permission check.
58. **Correct: A.**
- (A) **Correct.** Other roles then cannot spend the part reserved for the final decision.
 - (B) **Incorrect.** The assessment is a model call and incurs usage too.
 - (C) **Incorrect.** Earlier calls remain charged even after the patch reaches assessment.
 - (D) **Incorrect.** A late call still needs permission under the same run limit.
59. **Correct: A, C.**
- (A) **Correct.** Clearing a failed candidate must not erase charges incurred while producing it.
 - (B) **Incorrect.** Repeated attempts could then exceed the stated limit for the run.
 - (C) **Correct.** Every model role can create a charge, so leaving one out understates run spending.
 - (D) **Incorrect.** Failed calls and rejected attempts still consume resources.
60. **Correct: A, C, D.**
- (A) **Correct.** Holding these conditions fixed helps isolate the effect of how work is scheduled.
 - (B) **Incorrect.** A seed does not make different model inputs produce the same replies, and one trial cannot show variation.
 - (C) **Correct.** Keeping tools inside each fresh workspace blocks access to files left by earlier trials. Outcomes and costs show how and why runs stopped.
 - (D) **Correct.** Another model or budget could change both the trajectory and the result.